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| <b>Line Voltage-VF.CONCEPT</b> |
| <b>SFRS for Zefiro-VZES</b>    |

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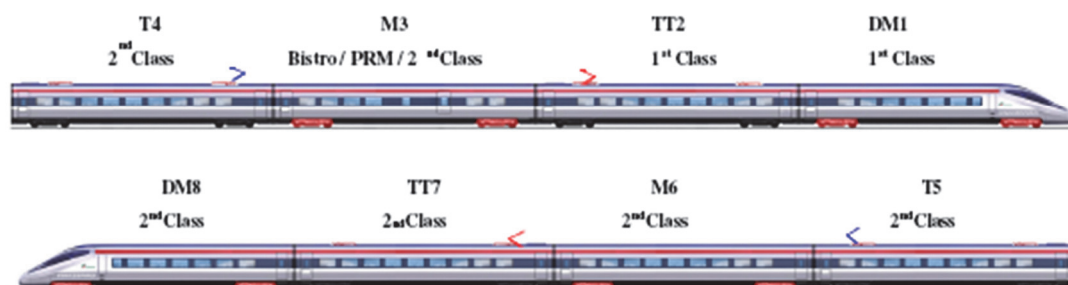
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## 1 Introduction

This document is a function specification for the control of the *Line Voltage system* for V300ZEFIRO for customer Trenitalia. (VZES Project).

### 1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

### 1.2 Definitions and abbreviations

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| AC          | Alternative Current                                                            |
| ADD         | Automatic Dropping Device                                                      |
| ATP         | Automatic Train Protection                                                     |
| CD          | Circuit Diagram                                                                |
| DC          | Direct Current                                                                 |
| DCU         | Drive Control Unit                                                             |
| DCUL        | Drive Control Unit / Line                                                      |
| DI-signal   | Digital Input signal                                                           |
| DO-signal   | Digital Output signal                                                          |
| ETCS System | European Train Control System                                                  |
| EVENT       | Diagnostics that will be stored in the Event database within TCMS SW.          |
| FMECA       | Failure Mode Effect and Criticality Analysis                                   |
| HW HMI      | Hardware Human Machine Interface, e.g. buttons, switches and indication lamps. |
| HV          | High Voltage                                                                   |
| HW          | Hardware                                                                       |
| IDU         | Interface Display Unit                                                         |
| KPS         | Relay energized by pressure switch of the pantograph                           |
| LCB         | Line Circuit Breaker                                                           |
| LCM         | Line Converter Module                                                          |
| LIM         | Line Interference Monitor                                                      |
| MCM         | Motor Converter Module                                                         |
| MVB         | Multifunction Vehicle Bus                                                      |
| N/A         | Not Applicable                                                                 |
| PCU         | Propulsion Control Unit                                                        |
| PCU         | Pantograph Control Unit                                                        |
| PPC         | Propulsion & Controls (Bombardier Division)                                    |
| PPU         | Pantograph Pneumatic Unit                                                      |
| SFRS        | System Function Requirement Specification                                      |

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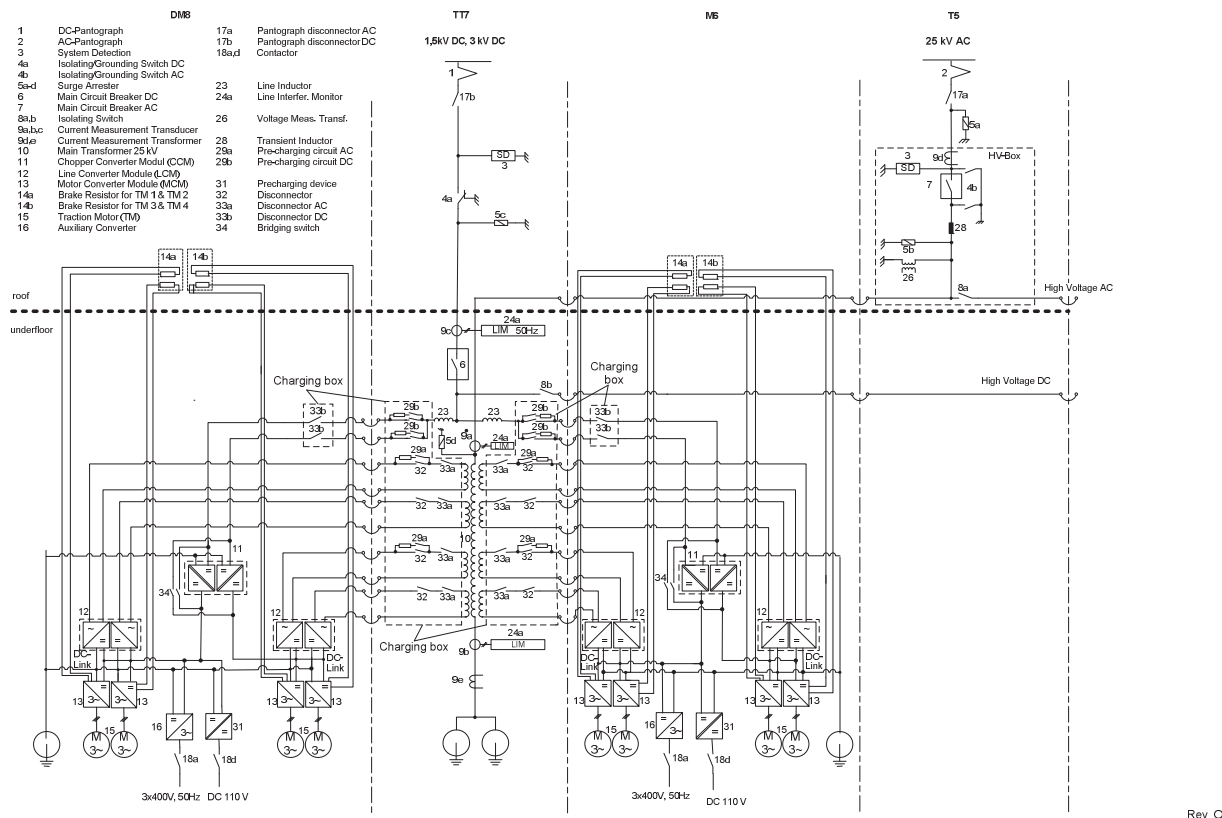
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|------|-------------------------------------|
| SIL  | Safety Integrity Level              |
| SIS  | Safety instrumented system          |
| SW   | Software                            |
| TBD  | To Be Defined                       |
| TCMS | Train Control and Management System |
| TRS  | Technical Requirement Specification |

All numbers in brackets refer to related numbers of system overview (see section 2.1).

## 2 System Overview

### 2.1 System Configuration

The architecture of the V300ZEFIRO energy distribution and traction/el. braking systems are designed with two Traction Units. Each traction unit consists of 4 coaches (half train), mirroring each other.



Rev. O

|                     |                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.670                                                                                                                                       |
| <b>Description</b>  | The architecture of the power supply system shall support the following four different line voltage systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC to 1.5 kV DC. |
| <b>Safety level</b> | N/A                                                                                                                                                                  |

The architecture of the power supply system supports the following four different line voltage systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC to 1.5 kV DC.

The line voltage system is selected by the train driver (see chapter 3.3.5 "system selection"). To identify the country of operation, the driver has to select the country manually.

The High voltage system is divided into two sections: AC high voltage section and DC high voltage section. Both parts are built-up redundantly, that means they consist of two pantographs for each catenary system and identical subsystems. Both line voltage subsystems are connected via separate high voltage cables, thus each of the two pantographs can supply the whole trainset simultaneously. All pantographs and the system detection are able to withstand 25 kV AC.

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### 3 Functional requirements

#### 3.1 Country code selection

##### 3.1.1 Function Design

###### 3.1.1.1 Normal condition

Add description what the Country code is used for functions.

###### 3.1.1.2 Failure condition

What kind of failure can it be with the country code selection

###### 3.1.1.3 Failure supervision

How can failures be detected and what action must the system/driver do?

##### 3.1.2 Requirements

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.772                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>  | <p>It shall be possible to select the country which the train will be operated in from the IDU. Possible counties are:</p> <ul style="list-style-type: none"> <li>– Italy, Austria, Belgium, France, Germany, Netherlands, Spain and Switzerland.</li> </ul> <p>The country selection shall be used for country specific configurations such as:</p> <ul style="list-style-type: none"> <li>– selection of wide or narrow pantograph</li> <li>– levels of line voltage supervision</li> <li>– levels of line current harmonics.</li> </ul> |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                     | Applicability Details | Derived to                  | User Interface | SIL   |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|----------------------|----------------------|---|-------|---|---------|---|---------|---|--------|---|---------|---|-------------|---|-------|---|-------------|--|--|---|-------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.775  | One of the following country-codes shall be used by TCMS depending the country in which the vehicle will operate:                                                                                                                                                                                                                                                                                                               | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 2 |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | <table><tr><td>Code</td><td>Country of operation</td></tr><tr><td>1</td><td>Italy</td></tr><tr><td>2</td><td>Austria</td></tr><tr><td>3</td><td>Belgium</td></tr><tr><td>4</td><td>France</td></tr><tr><td>5</td><td>Germany</td></tr><tr><td>6</td><td>Netherlands</td></tr><tr><td>7</td><td>Spain</td></tr><tr><td>8</td><td>Switzerland</td></tr><tr><td></td><td></td></tr><tr><td>0</td><td>not defined</td></tr></table> |                       |                             |                |       | Code                 | Country of operation | 1 | Italy | 2 | Austria | 3 | Belgium | 4 | France | 5 | Germany | 6 | Netherlands | 7 | Spain | 8 | Switzerland |  |  | 0 | not defined |
|                                     | Code                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |                             |                |       | Country of operation |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | Italy                |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | Austria              |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | Belgium              |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | France               |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 5                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | Germany              |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | Netherlands          |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     | 7                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |                             |                |       | Spain                |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
| 8                                   | Switzerland                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |                             |                |       |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                       |                             |                |       |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
| 0                                   | not defined                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |                             |                |       |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
| 2F_04.01.Zefiro-Europe.CONCEPT.965  | DELETED.                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 2 |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
| 2F_04.01.Zefiro-Europe.CONCEPT.1013 | The selected country code shall be sent to all PCU's in the train.                                                                                                                                                                                                                                                                                                                                                              | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 2 |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |
| 2F_04.01.Zefiro-Europe.CONCEPT.797  | In emergency operation mode (emergency operation mode trainline is energised) the latest selected/known country code shall be used and communicated to the PCU (LIM).                                                                                                                                                                                                                                                           | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 2 |                      |                      |   |       |   |         |   |         |   |        |   |         |   |             |   |       |   |             |  |  |   |             |

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## 3.2 Pantograph Control

### 3.2.1 Pantograph - Lifting

#### 3.2.1.1 *Function Design*

##### 3.2.1.1.1 Normal Condition

Lifting the pantograph is a precondition for collecting energy from catenary system.

Lifting can be performed automatically or manually. In any case, the driver selects a line voltage system (4-position selector switch) and switches "raise pantograph 1" or "raise pantograph 2".

Country Code is needed to identify pantograph which is needed depending on country of operation.

The identified pantograph are lifted up on request.

TCMS sends to PCU by IP bus which type of pantograph has to be lifted. TCMS also commands 2 different DOs in order to define which type of pantograph has to be lifted.

##### 3.2.1.1.2 Failure Condition

- 1) Pantograph doesn't lift up although an order was given.
- 2) Two pantographs of the selected type are lifting up. (unless for ice scraping)
- 3) Lifting of pantograph without order.
- 4) Wrong type of pantograph is lifting up.
- 5) Pantograph switch is in "up" position although related pantograph is lowered.

##### 3.2.1.1.3 Failure Supervision

- 1) If the pantograph is not raised after 35 seconds after the raise order and the main reservoir pressure is higher than 750 kPa, the pantograph is considered faulty.  
If no air pressure is detected at the pantograph 5 minutes after the raise order and the main reservoir pressure is less than or equal to 750 kPa, the pantograph is considered faulty.
- 2) Both pantographs will be ordered down.
- 3) Pantograph will be lowered, considered faulty and blocked additionally an event will be set.
- 4) system detection detects invalid line voltage system or country code does not fit to raised pantograph. In both cases all pantographs will be lowered and a failure message will be displayed.
- 5) TCMS ignores pantograph-switch and pantograph remains in lowered position. IDU indicates that switch has to be turned to "down" position.

#### 3.2.1.2 *Requirements*

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.619                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b>  | <p>The following sequence shall be implemented:</p> <ul style="list-style-type: none"> <li>* Driver selects the Country where the train is operated from the IDU.</li> <li>* Driver selects the line voltage system</li> <li>* The (not blocked or cut out) pantograph (one per 8 car train unit) for the chosen line voltage system shall be lifted when requested by the driver</li> <li>* Line voltage shall be detected and the chosen line voltage system verified</li> </ul> <p>If the detection and verification are successful, the closing of line circuit breakers shall be enabled. ELSE a diagnostic event shall be generated.</p> |
| <b>Safety level</b> | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Applicability Details | Derived to                                          | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.735  | With input from train driver (line voltage selector, see chapter 3.3.5 "System Selection") and depending on country of operation (country-code), TCMS shall select the needed type of pantograph (see chapter 3.2.5 "Pantograph selection").                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1099 | If a pantograph control unit (PCU) is faulty an event shall be set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1096 | In case all pantograph control units (PCU) are faulty, TCMS shall communicate directly to pneumatic control unit (PPU) which pantograph shall be ordered to raise.<br>The train speed shall be limited to 160 km/h.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1113 | The backup raise order to the pneumatic control unit (PPU) unit shall be connected to a TCMS DO. When the DO is energized the EV1 (EV2) order shall be notified to the PCU via IP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1114 | The major failure indication from PCU shall be connected to a TCMS DI.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1168 | If Major Fault from PCU has been detected in TCMS while the corresponding pantograph is raised, then "LCB emergency off" train line shall be immediately energized and the raising command shall be removed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1174 | Diagnosis of the contact: Major Fault (PCU)<br><br>During the start up phase, after self test of PCU, when the life signal from PCU is being received, this test begins and TCMS shall manage it:<br><ul style="list-style-type: none"> <li>TCMS reads the contact: Major Failure in closed position (i.e no failure inside PCU);</li> <li>TCMS sends for 6 s to PCU the command to open (just for test) the contact: Major Failure.</li> </ul><br>This test is being carried out only if the pantograph is in lowered position.<br>When the "Major Fault" contact test is on going, if the "Raise pantograph" switch is active, the raise order should be delayed to the end of the "Major Fault" contact test. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1175 | After self test, when PCU receives from TCMS the demand to carry out the test of the contact: "Major Fault", it has to follow exactly the request from TCMS in order to synchronize this test. PCU opens its own "Major Fault" contact for 2 s for test purpose.                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Derived Requirement   | 4S_04.01.02.-.Pantographs                           |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1185 | While TCMS is carrying out the test of Major Fault contact (i.e. TCMS is sending to PCU the request to open that contact):<br><ul style="list-style-type: none"> <li>TCMS reads the the contact: Major Failure in open position (i.e. it is not stuck);</li> <li>if the contact doesn't open during the test, at the end of this test an event shall be set (<b>Predisposition</b>: the affected pantograph shall be considered as "Faulty lowered")</li> </ul>                                                                                                                                                                                                                                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1237 | When major failure test is performed the TCMS shall rise an event for engineering of 'major fault test ongoing'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1236 | The major failure contact test has shall be done also when master reset is ordered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A   |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1097 | If the pantograph control unit rises the variable 'not rised on command' an event shall be raised to the driver | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1156 | When external power supply is active the pantograph shall not be allowed to raise.                              | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |

|                     |                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.767                                                                             |
| <b>Description</b>  | If the "Pantograph 1" switch is turned to "up" position the front pantograph in the train shall be raised. |
| <b>Safety level</b> | N/A                                                                                                        |

| Nr                                 | Description                                                                                                            | Applicability Details | Derived to                                          | User Interface | SIL |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.980 | The "up" position of the Pantograph 1 switch shall be connected to TCMS                                                | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.544 | When driver activates "Raise pantograph 1" switch, the front pantograph of the needed pantograph type shall be raised. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A |

|                     |                                                                                                           |
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| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.768                                                                            |
| <b>Description</b>  | If the "Pantograph 2" switch is turned to "up" position the rear pantograph in the train shall be raised. |
| <b>Safety level</b> | N/A                                                                                                       |

| Nr                                 | Description                                                                                                           | Applicability Details | Derived to                                          | User Interface | SIL |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.982 | The "up" position of the Pantograph 2 switch shall be connected to TCMS                                               | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.736 | When driver activates "Raise pantograph 2" switch, the rear pantograph of the needed pantograph type shall be raised. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A |

|                     |                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.685                                                                                              |
| <b>Description</b>  | For speeds greater than or equal to 300 km / h, the maximum number of pantographs raised shall be limited to two per train. |
| <b>Safety level</b> | N/A                                                                                                                         |

| Nr                                  | Description                                                                                                                                                                                                                                            | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.766  | When "Raise pantograph 2" switch is active and afterwards "Raise pantograph 1" switch will be activated, both pantographs of the selected pantograph type shall be raised. In multiple operation it is only allowed to raise one pantograph per train. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1058 | In case front pantograph is raised and selector for rear pantograph is activated, rear pantograph shall not be allowed to raise.                                                                                                                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1043 | An event shall be set if front pantograph is raised and selector for rear pantograph is activated stating that "Ice scraping is only allowed if rear pantograph is raised first".                                                                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.793  | In multiple operation, when driver activates "Raise pantograph 1" switch, in both coupled trains the front pantograph of the needed pantograph type shall be raised.                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1002 | In a multiple coupled train:<br>If pantograph 1 is selected to raise then the following actions shall be performed in priority order:<br>1) If both front pantographs are not ok then no pantograph shall be raised and an event shall be set for driver to change pantograph.<br>2) If front pantograph of first train is ok then front pantograph of first train shall be raised.<br>3) If front pantograph of second train is ok then front pantograph of second train shall be raised.<br>4) If front pantograph of first train is not ok and rear pantograph of first train is ok then rear pantograph of first train shall be raised.<br>5) If front pantograph of second train is not ok and rear pantograph of second train is ok then rear pantograph of second train shall be raised. | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.794  | In multiple operation, when driver activates "Raise pantograph 2" switch, in both coupled trains the rear pantograph of the needed pantograph type shall be raised.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1003 | In a multiple coupled train:<br>If pantograph 2 is selected to raise then the following actions shall be performed in priority order:<br>1) If both rear pantographs are not ok then no pantograph shall be raised and an event shall be set for driver to change pantograph.<br>2) If rear pantograph of first train is ok then rear pantograph of first train shall be raised.<br>3) If rear pantograph of second train is ok then rear pantograph of second train shall be raised.<br>4) If rear pantograph of first train is not ok and front pantograph of first train is ok then front pantograph of first train shall be raised.<br>5) If rear pantograph of second train is not ok and front pantograph of second train is ok then front pantograph of second train shall be raised.    | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1004 | If in DC operation rear pantograph of first train and front pantograph of second train are raised there shall be a speed restriction of 300 kph (fine tuning to be done later) and an event for driver to change pantograph "Speed restriction due to pantograph fault. Change pantograph if high speed on DC line is required"                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.768  | When an AC pantograph is not allowed to raise because there is already one up, this shall be indicated on IDU as an event.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.782  | When any local pantograph-switch is in 'up' position at the cab activation (Note: not during parking exit procedure) an event to the driver shall be set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.274  | The related LCB (6,7) and system isolating switch (4a only in DC line voltage system) and AC earthing switch (4b only in AC line voltage system) must be in open position to lift the pantograph.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.563  | When the pantograph is requested to raise, the pantograph disconnecter (17) shall be closed first. When the feedback that the disconnecter is closed is present, then the panto shall be ordered to raise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | N/A   |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1072 | The pantograph disconnecter shall close when ordered by TCMS DO.                                                                                                                                                                                                                                                                                       | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1159 | Pantograph control unit (PCU) commands to lift the selected pantograph, only if there is congruence between information coming via IP bus and signals from digital input.                                                                                                                                                                              | Derived Requirement | 4S_04.01.02.-<br>.Pantographs                           |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.281  | In order the keep the pantograph up the pantograph main control valve shall be kept open by energizing the solenoid.                                                                                                                                                                                                                                   | Derived Requirement | 4S_04.01.02.-<br>.Pantographs                           |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1173 | In case the IP communication between Pantograph Control Unit (PCU) and TCMS has been missed, PCU is not allowed to block and the pantograph mustn't be lowered.                                                                                                                                                                                        | Derived Requirement | 4S_04.01.02.-<br>.Pantographs                           |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1115 | The pantograph control unit (PCU) receives 2 different raise orders depending from the different type of pantograph. Each one shall be connected to a dedicated TCMS DO.                                                                                                                                                                               | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.542  | If the air pressure to the pantograph is low (pressure switch) the auxilliary compressor feeding the selected pantograph shall be ordered to start when the order to raise the pantograph is given.                                                                                                                                                    | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.543  | If the air pressure to the pantograph is low (pressure switch) AND the main recevoir pressure is <6 bar; the order to raise the pantograph shall be set when:<br>– pressure switch toggles to 'pressure available'<br><br>OR<br>– indipendently from the pressure switch after 6 minutes that the auxilliary compressor has been running continuously. | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.543  | In case the pantograph rise order (at the first rising) is delivered based on the condition: ' after 6 minutes that the auxilliary compressor has been running continuously' an event shall be set.                                                                                                                                                    | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.283  | The raise order needs to be continuously high to raise and keep the pantograph in raised position. A raise order <b>must not be cancelled within the first 2 seconds</b> after activating it, else the pantograph structure might be harmed.                                                                                                           | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1160 | When DC line voltage system is selected and parking mode is activated, the second pantograph of same type has to be raised by TCMS without command from driver, but no change in LCB position.<br>When AC line voltage system is selected and parking mode is activated, only the pantograph with the related LCB closed shall be kept raised.         | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.284  | The pantograph status, with following priority order:<br>– which pantos are "Cut out"<br>– which pantos are "Faulty raised"<br>– which pantos are "Faulty lowered"<br>– which pantos are "Raised"<br>– which pantos are "Lowered"<br>shall be displayed on the IDU                                                                                     | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          | IDU   | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.749  | When a pantograph is raised, the information which pantograph is raised shall be transmitted to propusion system to enable activation of related high voltage equipment.                                                                                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS<br>Software                          |       | SIL 0 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.737  | When pantograph is raised, present line voltage system shall be detected by system detection (see chapter 3.3.5.2 "System selection").                                                                                                                                                                                                                                     | Derived Requirement | 4S_04.-.Power Supply        |  | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1107 | For any pantograph configuration in multiple operation (AC or DC) a speed limitation has to be implemented. For now the limit will be equal to the max. test speed (final tuning will be done after first tests): <ul style="list-style-type: none"> <li>396 kph in AC 25kV</li> <li>330 kph in AC 15kV</li> <li>330 kph in DC 3kV</li> <li>242 kph in DC 1.5kV</li> </ul> | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |

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| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.304                                                                                                                                                                                                                  |
| <b>Description</b>  | To free the catenary line from ice it shall be possible to lift two pantographs per 8car train, however only one line circuit breaker is closed to supply the train with energy.<br>In this case it is necessary to foresee a speed limitation. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                                           |

To free the catenary line from ice it will be possible to lift two pantographs per 8-car train, however only one line circuit breaker is closed to supply the train with energy.  
In this case it is necessary to foresee a speed limitation.

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               | Applicability Details | Derived to                  | User Interface | SIL   |
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| 2F_04.01.Zefiro-Europe.CONCEPT.783  | For ice scraping it shall be allowed to turn both pantograph switches to "up" position when the train is operating in Italy (country-code = 1) or France or Spain.                                                                                                                                                                                                                                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1238 | When external temperature is <3°C <b>AND</b><br>train is running (V> 3km/h) <b>AND</b><br>the line selector is on 3kV <b>OR</b> 1,5kV<br>a message 'risk of frozen catenary line' shall be raised to the driver after a time delay of 10 sec.<br><br>the message shall be reset in the following cases:<br>Ice scraping mode has been activated <b>OR</b><br>line selector set on AC mode <b>OR</b><br>external temperature is >=5°C for more than 10 sec | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.784  | When both pantographs are raised for ice-scraping, the enable signal to close the LCB shall only be given to LCB which is related to the rear pantograph!                                                                                                                                                                                                                                                                                                 | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.559  | For ice scraping the pantograph closest to the active cab shall be used.<br>NOTE! No ice scraping pantograph shall be used in a coupled train nor in backup mode.                                                                                                                                                                                                                                                                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.560  | Ice scraping mode shall only be allowed in Italy, France and Spain.                                                                                                                                                                                                                                                                                                                                                                                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

When a second pantograph is raised for ice scraping, it is in responsibility of the driver to limit the maximum speed at high speed lines to 200 km/h.

| Nr                                 | Description                                                                                                                         | Applicability Details | Derived to                  | User Interface | SIL   |
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| 2F_04.01.Zefiro-Europe.CONCEPT.548 | If two DC pantographs is raised for ice scraping, a warning message shall be displayed on the IDU.                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.484 | The status of the ice scraping pantograph shall be displayed on the IDU in the same way as for a normal power collecting pantograph | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 0 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.486 | If a second pantograph is raised (for ice scraping) then the LCB and the System Isolation Switch (only in DC) connected to the second pantograph raised shall be interlocked.             | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |  | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.748 | If a second pantograph is raised (for ice scraping) then the LCB and the System Isolation Switch (only in DC) connected to the second pantograph raised shall be interlocked by software. | Derived Requirement | 4S_04.-.Power Supply                                    |  | SIL 0 |

## 3.2.2 Pantograph - Lowering

### 3.2.2.1 Function Design

#### 3.2.2.1.1 Normal Condition

Lowering of pantograph, the Pantograph will stop collecting line current.

Pantograph will be lowered when line voltage system is being de-activated:

- by switch “Pantograph down” OR
- by drivers urgency (“Emergency push button”) push button OR
- by automatic device due to system change OR
- in case of critical failures (Pantograph control unit output “pantograph internal major failure” in case of Pantograph control unit fault or difference in pressure control)
- at cab deactivation (no parking mode active)

All line circuit breakers are ordered to open before lowering the pantograph except in emergency cases.

#### 3.2.2.1.2 Failure Condition

- 1) Pantograph doesn't lower on order.
- 2) Pantograph is lowered unintentionally.

#### 3.2.2.1.3 Failure Supervision

- 1) Manual lowering via panto-cock.
- 2) The faulty pantograph will be blocked.

### 3.2.2.2 Requirements

|                     |                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.266                                                                                                                                                                                                                                                                                                                                           |
| <b>Description</b>  | Pantograph shall be lowered when line voltage system shall be de-activated <ul style="list-style-type: none"> <li>– by switching the selector switch to the down position OR</li> <li>– by drivers emergency push button OR</li> <li>– in case of failures.</li> </ul> All line circuit breakers shall be open before lowering the pantograph except in emergency cases. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                                                                                                                                                                    |

|                     |                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.769                                                                                |
| <b>Description</b>  | If the "Pantograph 1" switch is turned to "down" position the front pantograph in the train shall be lowered. |
| <b>Safety level</b> | N/A                                                                                                           |

|                     |                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.770                                                                               |
| <b>Description</b>  | If the "Pantograph 2" switch is turned to "down" position the rear pantograph in the train shall be lowered. |
| <b>Safety level</b> | N/A                                                                                                          |

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| Nr                                  | Description                                                                                                                                                                                                                                     | Applicability Details | Derived to                                              | User Interface | SIL   |
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| 2F_04.01.Zefiro-Europe.CONCEPT.984  | The "down" position of the Pantograph 1 switch shall be connected to TCMS                                                                                                                                                                       | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.983  | The "down" position of the Pantograph 2 switch shall be connected to TCMS                                                                                                                                                                       | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.968  | When driver activates "Lower pantograph 1" switch, the front pantograph of the needed pantograph type shall be lowered.                                                                                                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.969  | When driver activates "Lower pantograph 2" switch, the rear pantograph of the needed pantograph type shall be lowered.                                                                                                                          | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.561  | When panto down order is given, the pantograph shall lower as soon as the LCB has opened or latest 3 seconds after the order was given.                                                                                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1219 | When a panto lowering order is given, a new rise command to the same pantograph shall be ignored until the PCU delivers the information of panto lowered (Ipanpos = 0).                                                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1220 | In case a panto rise order is requested from the driver while the pantograph moving signal is set (during the lowering phase) an advisory should be displayed to driver asking to wait for the lowering phase to finish.                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1225 | When the pantograph is lowered (Ipanpos = 0), and the driver has tried to re-rise it during the lowering phase, an event shall be set to tell the driver that the panto is risable again.                                                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.370  | The pantograph shall not lower, except for specific emergency situations (stated in other requirements) if the LCB is closed or line current is detected.                                                                                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.565  | The pantograph disconnecter shall be ordered to open after the pantograph has been lowered or at latest 10 seconds after the pantograph down order was given.                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1073 | The pantograph disconnecter shall open when ordered by TCMS DO.                                                                                                                                                                                 | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.77   | In order to lower the pantograph the pantograph main control valve shall be switched over to drain the air pipes by removing the power to the solenoid.                                                                                         | Derived Requirement   | 4S_04.01.02.-<br>.Pantographs                           |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.289  | When lowering a DC pantograph the system isolation switch shall open when the pantograph is in down position.                                                                                                                                   | Derived Requirement   | 4S_04.-.Power Supply                                    |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.562  | The pantograph shall be order to lower 500ms after an emergency stop button is pressed, regardless of the position of the LCB.                                                                                                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1098 | The pantograph shall lower, using the fast lowering valve (panto fast down train line), 500ms after an emergency stop button is pressed, regardless of the position of the LCB.                                                                 | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.994  | Any raised pantograph shall be lowered when requested from the PCU.                                                                                                                                                                             | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1008 | If the cab is deactivated without entering the parking mode all pantographs shall be lowered.                                                                                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1153 | In case pantograph is lowered without order (signal from PCU) the LCB shall be opened by activation of "LCB emergency open" trainline, the pantograph order shall be removed and an alarm shall be set accordign to the diagnosys from the PCU. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | SIL 0 |

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|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|--|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.1165 | <p>In case Pantograph Control unit (PCU) detects incongruence between HW and SW 'raising signal' (= SW rise order is missing OR HW rise order is missing), the following actions shall be carried out.</p> <p>After 2 s from inchoerence detection:</p> <ul style="list-style-type: none"> <li>– <b>Predisposition:</b> LCB shall be ordered to open (TCMS via SW);</li> <li>– <b>Predisposition:</b> the pantograph shall be lowered and blocked</li> <li>– <b>Predisposition:</b> an event shall be set to the driver (Change pantograph required)</li> <li>– the related fault event shall be set to the maintainer (incoherence between HW and SW orders or vice-versa)</li> </ul> <p>After 16 s from inchoerence detection (if the pantograph has not been lowered):</p> <ul style="list-style-type: none"> <li>– <b>Predisposition:</b> TCMS shall energize "Panto Fast down " train line;</li> </ul> | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1172 | <p>If the Pantograph Control Unit (of the panto in use) communication to the TCMS is NOT OK the following sequence shall be ordered:</p> <ul style="list-style-type: none"> <li>– LCB emergency open trainline shall be energized;</li> <li>– after 500 ms, Panto Fast down trainline shall be energized and and the raising command shall be removed;</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | N/A |

### 3.2.3 Automatic Dropping Device (ADD)

#### 3.2.3.1 Function Design

##### 3.2.3.1.1 Normal Condition

Lowering pantograph in case of collector head failure.

In order to have information about strips wearing there are two tubes inside each contact strip that are connected with pneumatic system, the tube at the highest level provides information about the wear of contact strips. The second tube at the lowest level activates the automatic lowering of pantograph (ADD).

When the ADD detects damages of contact strips it will drop the pantograph immediately.

In the event of collector head failure all pantographs of the train will be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs.

##### 3.2.3.1.2 Failure Condition

- 1) Strip is faulty but not detected by ADD.
- 2) Strip is not faulty but failure is detected by ADD.

##### 3.2.3.1.3 Failure Supervision

- 1) No error-handling possible.
- 2) A faulty ADD-release will be handled as a normal ADD-release.

#### 3.2.3.2 Requirements

|                     |                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.445                                                                   |
| <b>Description</b>  | The ADD shall detect damages of contact strips (rupture, abnormal wear) and insulated end horns. |
| <b>Safety level</b> | SIL 2                                                                                            |

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| Nr                                 | Description                                                                                     | Applicability Details | Derived to                    | User Interface | SIL   |
|------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.332 | The ADD shall drop the pantograph immediately if any damage to the contact strips are detected. | Derived Requirement   | 4S_04.01.02.-<br>.Pantographs |                | SIL 2 |

|                     |                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.270                                                                                                                                               |
| <b>Description</b>  | In the event of collector head failure all pantographs of the train shall be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs. |
| <b>Safety level</b> | SIL 2                                                                                                                                                                        |

| Nr                                  | Description                                                                                                                                                        | Applicability Details | Derived to                                              | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1061 | When wear detection is activated, the Pantograph control unit (PCU) shall communicate to TCMS that a contact strip is worn.                                        | Derived Requirement   | 4S_04.01.02.-<br>.Pantographs                           |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1062 | If wear detection is tripped an EVENT shall be set.                                                                                                                | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1063 | If wear detection is tripped a counter shall be activated to measure the running distance with the worn pantograph in use.                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1119 | If worn pantograph is in use, TCMS shall send to PCU (pantograph control unit) the process data "Wear detection blocked".                                          | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1064 | The counter shall be allowed to reset to 0 by maintenance people when worn contact strip has been changed.                                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.82   | If ADD is tripped the status for the pantograph shown on the IDU shall be "faulty".                                                                                | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | IDU            | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.294  | If ADD is tripped an event shall be set.                                                                                                                           | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.295  | If ADD is tripped the LCBs shall open immediately.                                                                                                                 | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.747  | If ADD is tripped the LCBs shall open immediately.                                                                                                                 | Derived Requirement   | 4S_04.-.Power Supply                                    |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.371  | If ADD is tripped the LCBs shall be requested to open immediately.                                                                                                 | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1178 | If ADD is tripped and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set                             | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1169 | In case ADD has been detected, via the KPD relay, the raising command from TCMS has to be removed                                                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1170 | Pantograph Control unit (PCU) resets the signal: "ADD tripped" after 30 s that the pantograph has been folded.                                                     | Derived Requirement   | 4S_04.01.02.-<br>.Pantographs                           |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1171 | The condition: "ADD tripped" is being reset in TCMS as soon as the signal (relay KPD) has gone to 0                                                                | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1067 | In case of ADD tripped the pantograph has to be declared as faulty and the panto has to be blocked. The panto will be operational again only after a master reset. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1110 | A block of pantograph due to ADD-trip shall be allowed to reset only by maintenance people.                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1068 | If a faulty pantograph is blocked (due to ADD-trip), an EVENT shall be set.                                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.297  | If ADD trips and train speed is > 0 km/h every pantograph in the whole train shall be lowered immediately via the "panto fast down" trainline.                     | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1166 | If "Panto Fast down" trainline is energized, TCMS sends this information to Pantograph Control Unit (PCU) via IP                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | N/A   |

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| 2F_04.01.Zefiro-Europe.CONCEPT.296  | If ADD trips and train speed is > 0 km/h every LCB shall requested to be opened and all pantographs in the whole train shall be lowered.                                                                                                                                                                                                                     | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.372  | The "fast panto down" trainline shall be set for 30 seconds when a ADD trip occurs.                                                                                                                                                                                                                                                                          | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1094 | When pantograph raise order is given, activation of "fast panto down" trainline shall be delayed by n seconds to avoid that all pantographs will be lowered immediately.                                                                                                                                                                                     | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.298  | If the carbon strip is faulty the pneumatic control loop for the pantograph shall be drained.                                                                                                                                                                                                                                                                | Derived Requirement | 4S_04.01.02.-.Pantographs                           |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.300  | Insulating distance shall be reached immediately (worst case after 1s).                                                                                                                                                                                                                                                                                      | Derived Requirement | 4S_04.01.02.-.Pantographs                           |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.301  | After all pantographs have been dropped by ADD activating the "panto fast down" trainline, a non faulty pantograph shall be possible to select by the driver.                                                                                                                                                                                                | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1176 | To avoid false diagnosis, PCU has to acquire the status of EV3 (inside PPU) and consequently to mask the diagnostic signal "ADD tripped" if EV3 has been energized                                                                                                                                                                                           | Derived Requirement | 4S_04.01.02.-.Pantographs                           |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1157 | In case of failure of relays in ADD detection circuitry (unavailability for LCB opening and panto fast down train lines activation), the raising command for the pantograph shall be blocked and an EVENT shall be set.<br>Note: this diagnosis has to be masked in case of "Panto Fast down" train line has been energized or any A-Key has been activated. | Derived Requirement | 4S_09.03.05.-.TCMS Software                         | EVENT | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1158 | Even though the pantograph was blocked due to failure recognised in ADD detection circuitry, in emergency operation mode the pantograph raising command shall be allowed.                                                                                                                                                                                    | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |

|                     |                                                                                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.794                                                                                                                                                                                                                                                                                           |
| <b>Description</b>  | In case of unavailability of both pantographs of the available system due to ADD intervention (or failure), through a sealed switch in driver's cab the driver, under his responsibility, can allow to raise the pantograph and move the train at reduced speed in order to reach a siding track to clear the main line. |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                                                                                                    |

| Nr                                  | Description                                                                                                                                                                                                                                                                                            | Applicability Details | Derived to                                          | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1291 | GIVEN driver has switched the ADD bypass switch to AC or DC position AND the TCMS has accordingly activated the related DO of the ADD bypass solenoid valve<br>THEN<br>this solenoid valve prevent the leakage of air pressure due to any damage of contact strip on those pantograph types (AC or DC) | Derived Requirement   | 4S_04.01.02.-.Pantographs                           |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1292 | GIVEN the driver has switched the ADD bypass switch to DC position<br>THEN CTC shall forward the status DC ADD bypass active to TCMS via safe MIO (DI)<br>Note:- DC ADD bypass means the ADD function of the DC pantograph is bypassed                                                                 | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1293 | GIVEN the driver has switched the ADD bypass switch to AC position<br>THEN CTC shall forward the status AC ADD bypass active to TCMS via safe MIO (DI)<br>Note:- AC ADD bypass means the ADD function of the AC pantograph is bypassed                                                                 | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1339 | ADD bypass switch is in "0", "DC" and "AC" according concept 1340                                                                                                                                                                                                                                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | SIL 2 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1341 | Event related to ADD bypass switch according concept 1340                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software                             | EVENT    | SIL 0                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
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| 2F_04.01.Zefiro-Europe.CONCEPT.1340 | <table><thead><tr><th colspan="5">INPUT</th><th colspan="4">OUTPUT</th></tr><tr><th>MNO safe DC1</th><th>MNO safe DC2</th><th>MNO safe AC1</th><th>MNO safe AC2</th><th>Validity</th><th>Status ADD Bypass Switch</th><th>SET Event ADD Bypass Switch</th><th>RESET Event ADD Bypass Switch</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>DC</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>AC</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>X</td><td>X</td><td>X</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>X</td><td>X</td><td>X</td><td>0</td><td>1</td><td>0</td></tr><tr><td>X</td><td>X</td><td>1</td><td>0</td><td>X</td><td>0</td><td>1</td><td>0</td></tr><tr><td>X</td><td>X</td><td>0</td><td>1</td><td>X</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>X</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr></tbody></table> <p>Validity = 1 if all inputs (DC1, DC2, AC1, AC2) are valid<br/>X = 1 OR 0</p> | INPUT               |                                                         |          |                          |                             | OUTPUT                        |  |  |  | MNO safe DC1 | MNO safe DC2 | MNO safe AC1 | MNO safe AC2 | Validity | Status ADD Bypass Switch | SET Event ADD Bypass Switch | RESET Event ADD Bypass Switch | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | DC | 0 | 1 | 0 | 0 | 1 | 1 | 1 | AC | 0 | 1 | 1 | 0 | X | X | X | 0 | 1 | 0 | 0 | 1 | X | X | X | 0 | 1 | 0 | X | X | 1 | 0 | X | 0 | 1 | 0 | X | X | 0 | 1 | X | 0 | 1 | 0 | 1 | 1 | 1 | 1 | X | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 2 |
| INPUT                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |                                                         |          | OUTPUT                   |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| MNO safe DC1                        | MNO safe DC2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MNO safe AC1        | MNO safe AC2                                            | Validity | Status ADD Bypass Switch | SET Event ADD Bypass Switch | RESET Event ADD Bypass Switch |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 0                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                   | 0                                                       | 1        | 0                        | 0                           | 1                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 1                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                   | 0                                                       | 1        | DC                       | 0                           | 1                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 0                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                   | 1                                                       | 1        | AC                       | 0                           | 1                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 1                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | X                   | X                                                       | X        | 0                        | 1                           | 0                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 0                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | X                   | X                                                       | X        | 0                        | 1                           | 0                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| X                                   | X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                   | 0                                                       | X        | 0                        | 1                           | 0                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| X                                   | X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                   | 1                                                       | X        | 0                        | 1                           | 0                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 1                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                   | 1                                                       | X        | 0                        | 1                           | 0                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 0                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                   | 0                                                       | 0        | 0                        | 1                           | 0                             |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1295 | GIVEN<br>the ADD bypass switch in active driver's cab is in position "DC" OR<br>the ADD bypass switch in active driver's cab is in position "AC"<br>WHEN<br>vehicle is at standstill<br>THEN<br>an event shall be set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Derived Requirement | 4S_09.03.05.-.TCMS Software                             | EVENT    | SIL 0                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1296 | GIVEN<br>the ADD bypass switch in active driver's cab is in position "DC"<br>WHEN<br>vehicle is at standstill<br>THEN<br>the solenoid valve for ADD bypass on both TT cars shall be energized by safe DOs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software                             |          | SIL 2                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1297 | GIVEN<br>the ADD bypass switch in active driver's cab is in position "AC"<br>WHEN<br>vehicle is at standstill<br>THEN<br>the solenoid valve for ADD bypass on both T cars shall be energized by safe DOs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Derived Requirement | 4S_09.03.05.-.TCMS Software                             |          | SIL 2                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1298 | GIVEN<br>local cab is not active<br>WHEN<br>the ADD bypass switch in local cab is in position "DC" OR<br>the ADD bypass switch in local cab is in position "AC"<br>THEN<br>an event shall be set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement | 4S_09.03.05.-.TCMS Software                             | EVENT    | SIL 0                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1307 | GIVEN<br>the event as for req#1298 has been set in this local cab<br>WHEN<br>the ADD bypass switch in local cab is not in position "DC" AND<br>the ADD bypass switch in local cab is not in position "AC"<br>THEN<br>the event as for req#1298 for this local cab shall be reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement | 4S_09.03.05.-.TCMS Software                             | EVENT    | SIL 0                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1300 | GIVEN driver has switched the ADD bypass switch to AC or DC position AND<br>the TCMS has accordingly activated the safe DOs to energize in parallel, via diode, the KPS relay<br>THEN<br>the KPS relay usually fed by pressure switch (PS3 / PS4) is in position of ADD intervention not activated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |          | SIL 2                    |                             |                               |  |  |  |              |              |              |              |          |                          |                             |                               |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                     |                             |  |       |

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|                                     |                                                                                                                                                                                                                                                                                                                                                                   |                     |                             |  |       |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|--|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1301 | <i>The feedback from pressure switch to indicate the ADD intervention (PS3 / PS4) shall be suppressed by TCMS in the following way:</i><br>GIVEN<br>the ADD bypass switch in active driver's cab is in position "DC"<br>THEN<br>TCMS shall activate the safe DOs in order to energize the KPS relay related to ADD pressure switch (PS3 / PS4) , on both TT cars. | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1302 | <i>The feedback from pressure switch to indicate the ADD intervention (PS3 / PS4) shall be suppressed by TCMS in the following way:</i><br>GIVEN<br>the ADD bypass switch in active driver's cab is in position "AC"<br>THEN<br>TCMS shall activate the safe DOs in order to energize the KPS relay related to ADD pressure switch (PS3 / PS4) , on both T cars.  | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1303 | GIVEN<br>the ADD bypass switch in active driver's cab is in position "DC"<br>THEN<br>TCMS shall ignore any ADD intervention received from DC pantographs, in order to prevent any action which would bring to isolate the corresponding pantograph.                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1304 | GIVEN<br>the ADD bypass switch in active driver's cab is in position "AC"<br>THEN<br>TCMS shall ignore any ADD intervention received from AC pantographs, in order to prevent any action which would bring to isolate the corresponding pantograph.                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1305 | GIVEN<br>pantograph has been declared as faulty and is blocked due to ADD intervention<br>WHEN<br>ADD bypass for that type of pantograph is active<br>THEN<br>panto blocking shall be temporarily suppressed and can be operational again                                                                                                                         | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1306 | GIVEN<br>panto blocking has been temporarily suppressed<br>WHEN<br>ADD bypass for that type of pantograph is no longer active<br>THEN<br>the pantograph remains blocked                                                                                                                                                                                           | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1333 | GIVEN<br>ADD bypass switch in active driver's cab is in position "DC" OR<br>ADD bypass switch in active driver's cab is in position "AC"<br>THEN<br>Panto Control Unit shall receive information of "ADD Bypass Activated"                                                                                                                                        | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |

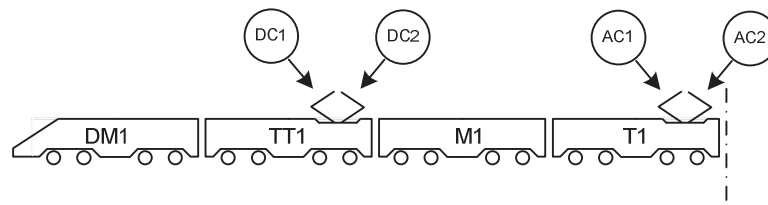
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## 3.2.4 Pantograph - Interlock

### 3.2.4.1 Function Design

#### 3.2.4.1.1 Normal Condition

Faulty pantographs are blocked automatically. The driver can block a pantograph manually to prevent switching or if it is defect without automatic detection.



#### 3.2.4.1.2 Failure Condition

- 1) Unidentified lifting of pantograph.

#### 3.2.4.1.3 Failure Supervision

- 1) Pantograph will be lowered and blocked.

### 3.2.4.2 Requirements

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.275                           |
| <b>Description</b>  | The driver shall be able to manually block a pantograph. |
| <b>Safety level</b> | N/A                                                      |

| Nr                                 | Description                                                                                                                     | Applicability Details | Derived to                  | User Interface | SIL |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.87  | The driver shall be able to manually cut out a pantograph from selection                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.303 | If the pantograph is manually cut out an event shall be set as long as the pantograph is blocked.                               | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.571 | If a raised pantograph is manually cut out the LCB shall be requested to open and the pantograph shall lower.                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.304 | It shall not be possible to remove a manual cut out of the pantograph as long as any pantograph is raised.                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.305 | If a pantograph is blocked it shall always be considered down regardless of the pressure sensor indicates low or high pressure. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.306 | If a pantograph is blocked the raise order shall be constant low.                                                               | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.621                      |
| <b>Description</b>  | A faulty pantograph shall automatically be blocked. |
| <b>Safety level</b> | SIL 0                                               |

| Nr                                 | Description                                                                                                                                      | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.309 | In case the pantograph does not lower on command the diagnosys of the PCU shall rise an alarm to the driver and the pantograph shall be blocked. | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | SIL 0 |

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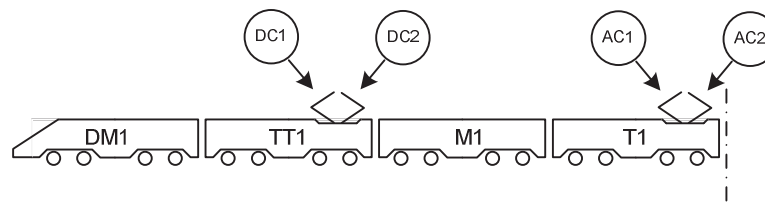
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| 2F_04.01.Zefiro-Europe.CONCEPT.310  | If the pantograph is in moving status (checking the pantograph position variable on IP) for t>20s an event shall be set.                                                                                                                                                                           | Derived Requirement | 4S_09.03.05.-.TCMS Software                         | EVENT | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.566  | If the pantograph disconnecter does not open when ordered the pantographs connected to the same LCB (DC1-DC2 or AC1-AC2) shall be permanently blocked and an event shall be set.                                                                                                                   | Derived Requirement | 4S_09.03.05.-.TCMS Software                         | EVENT | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.567  | If the pantograph disconnecter does not close when ordered the pantograph shall be lowered, the pantographs connected to the same LCB (DC1-DC2 or AC1-AC2) shall be permanently blocked and an event shall be set.                                                                                 | Derived Requirement | 4S_09.03.05.-.TCMS Software                         | EVENT | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1103 | If the pantograph disconnecter opens without order ("closed" feedback disappears) while the pantograph is raised, the LCB shall be opened, the pantograph shall be lowered, the pantographs connected to the same LCB (DC1-DC2 or AC1-AC2) shall be permanently blocked and an event shall be set. | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1074 | Both the positions open and closed of the pantograph disconnecter shall be connected to TCMS DI.                                                                                                                                                                                                   | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits |       | N/A   |

## 3.2.5 Pantograph Selection

### 3.2.5.1 Function Design

#### 3.2.5.1.1 Normal Condition

Pantographs of different type (AC and DC) and different width (1450mm, 1600mm and 1950mm) is used. Depending of the country the train operates in different width is required. Therefore, a country code to identify the current country of operations selected by the driver.



DC1 - 1450mm for DC-operation in Italy

DC2 - 1950mm for DC-operation in all countries but Italy

AC1- 1450mm or 1600mm for AC-operation in Italy, France, Belgium, Switzerland and Spain.

\* 1450mm for operation in a corridor including Italy and Switzerland.

\* 1600mm for operation in a corridor including Italy and Spain.

\* For Belgium and France both 1450mm and 1600mm can be used.

AC2 - 1950mm for AC-operation in Germany, Austria, Netherlands and Spain.

#### 3.2.5.1.2 Failure Condition

- 1) Wrong pantograph type is ordered to lift up.
- 2) More than one pantograph gets raise order.

#### 3.2.5.1.3 Failure Supervision

- 1) Driver supervises pantograph selection.
- 2) With exception of raising a second pantograph (ice scraping) all LCBs will open immediately and all pantographs will be lowered, if - in one 8 car train - more than one pantograph is raised.

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### 3.2.5.2 Requirements

|                     |                                                                                                                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.286                                                                                                                                                                                                                                                                              |
| <b>Description</b>  | The 4 pantographs of one half-train (4 cars), which are used for AC voltage of 25 kV 50 Hz or 15 kV 16.7 Hz and DC voltage of 3 kV or 1,5 kV, shall be interlocked against each other. Switch over between the systems shall be performed automatically onboard the train set while the train is in motion. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                                                                                                       |

Only one pantograph on a full train set (8-cars) is selected for raising. The only situation when two pantograph selected is when one is used for ice scraping, see section 3.2.1

| Nr                                  | Description                                                                                                                                          | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.786  | When voltage system AC 25kV or AC 15kV is selected and the train operates in Spain, the <u>narrow</u> AC-pantograph (AC1, 1600mm) shall be selected. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1329 | When voltage system DC 3kV is selected and the train operates in Spain, the <u>wider</u> DC-pantograph (DC2) shall be selected.                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.496  | If the LCB connected to the selected pantograph is blocked, it shall be indicated on IDU and the driver will have to raise the other pantograph.     | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1095 | When one of the following countries is selected all DC pantographs shall be blocked: Germany, Austria, Switzerland.                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

|                     |                                                                                                                                                                                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.325                                                                                                                                                                                                                                                                                                |
| <b>Description</b>  | It shall be possible, through appropriate forcing procedures, to collect energy from any type of catenary by using the other AC pantograph of the trainset for AC lines or the other DC pantograph of the trainset for DC lines. This is limited for special purpose (like testing) under specific and supervised conditions. |
| <b>Safety level</b> | N/A                                                                                                                                                                                                                                                                                                                           |

| Nr                                 | Description                                                                                                                                                                                                                                        | Applicability Details | Derived to                  | User Interface | SIL |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.216 | It shall be possible to control each pantograph individually by test parameters in the software. These parameters shall only be possible to be manipulated by connecting directly to the CCU with appropriate commissioning tools such as DCUTerm. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |

## 3.2.6 Pantograph - Control of Static Contact Force

### 3.2.6.1 Function Design

#### 3.2.6.1.1 Normal Condition

The static force of the raised pantograph will be controlled depending on speed, travel direction, position of pantograph and voltage system.

#### 3.2.6.1.2 Failure Condition

- 1) Control system is not working correctly.
- 2) Ethernet link is missing.

#### 3.2.6.1.3 Failure Supervision

- 1) Switching to other pantographs.
- 2) Switching to other pantographs.

|                                    |                         |                         |                     |                        |
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### 3.2.6.2 Requirements

|                     |                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.454                                                                         |
| <b>Description</b>  | The pressure shall be a function of voltage system, speed, travel direction and position of pantograph |
| <b>Safety level</b> | N/A                                                                                                    |

The pressure is a function of voltage system, speed, travel direction and position of pantograph

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Applicability Details | Derived to                    | User Interface | SIL   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.97   | The static contact force of the pantograph shall be regulated using the following parameters <ul style="list-style-type: none"> <li>– Voltage system (AC or DC)</li> <li>– Train speed</li> <li>– Travel direction (Backwards and Forward)</li> <li>– Position of all raised pantographs in a train (counting from 1, closest to active cab<br/>Note: for a coupled consist the numbering continues)</li> <li>– applicability of ice-scraping mode</li> </ul> | Derived Requirement   | 4S_04.01.02.-<br>.Pantographs |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1111 | TCMS shall provide following parameters to pantograph control unit (PCU): <ul style="list-style-type: none"> <li>– Voltage system (AC or DC)</li> <li>– Train speed</li> <li>– Travel direction (Backwards and Forward)</li> <li>– Position of all raised pantographs in a train (counting from 1, closest to active cab<br/>Note: for a coupled consist the numbering continues)</li> <li>– applicability of ice-scraping mode</li> </ul>                    | Derived Requirement   | 4S_09.03.05.-.TCMS Software   |                | SIL 0 |

## 3.3 Supervision of Line Voltage and Line Current

### 3.3.1 Line Voltage Supervision

#### 3.3.1.1 Function Design

##### 3.3.1.1.1 Normal Condition

The supervision of the Line Voltage is done by the voltage sensors connected to the propulsion equipment. If the line voltage detection indicates a wrong line voltage system or a too high or too low Line Voltage, the active LCB will be opened respectively not closed.

Precondition: A country code is selected to identify the country of operation.

##### 3.3.1.1.2 Failure Condition

- 1) No signal from system detection AC or DC (3).
- 2) Incorrect measurement (implausible voltage value) in system detection AC or DC (3).
- 3) No signal from AC voltage transformer (26)
- 4) Incorrect measurement (implausible voltage value) in AC voltage detection (26).

##### 3.3.1.1.3 Failure Supervision

- 1) Driver has to change pantograph.
- 2) Driver has to change pantograph.
- 3) The error handling has to isolate the faulty half-train by opening of roof line disconnectors 8a, refer to section 3.4.3.
- 4) The error handling has to isolate the faulty half-train by opening of roof line disconnectors 8a, refer to section 3.4.3.

|                                    |                         |                         |                     |                        |
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### 3.3.1.2 Requirements

|                     |                                                                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.458                                                                                                                        |
| <b>Description</b>  | Line voltage shall be supervised if it is within acceptable range to allow the line circuit breakers to close and to release full or reduced traction |
| <b>Safety level</b> | SIL 1                                                                                                                                                 |

If "AC system detection" (inside HV-Box, T4 & T5 car) detects AC (50Hz or 16.7Hz) contact wire voltage, the voltage of AC-system will be measured in addition (15 kV or 25 kV).

If DC system detection (TT2 & TT7 car) detects DC contact wire voltage, the voltage of DC-system will be measured in addition (1.5 kV or 3 kV).

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.222  | The AC-LCB shall be allowed to close when the line voltage detection AC 50 Hz is $\geq 17.5$ kV and $\leq 29.0$ kV or line voltage detection AC 16.7 Hz is $\geq 11.0$ kV and $\leq 18.0$ kV.                                                                                                                                                                                                                                                                                                                                                                                                                          | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1216 | The LCB closing request has to be delivered to Propulsion only if:<br><ul style="list-style-type: none"> <li>the line voltage detection AC 50 Hz is <math>\geq 17.5</math> kV and <math>\leq 29.0</math> kV or line voltage detection AC 16.7 Hz is <math>\geq 11.0</math> kV and <math>\leq 18.0</math> kV</li> <li>the line voltage detection DC 3kV is <math>\geq 1.975</math> kV and <math>\leq 3.9</math> kV (operation in Italy: <math>\leq 4.2</math> kV, operation in Belgium: <math>\leq 3.8</math> kV) or line voltage detection DC 1.5kV <math>\geq 1.0</math> kV and <math>&lt; 1.975</math> kV</li> </ul> | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1217 | In case of requests to close the LCB with a measured line voltage not within the limits above an advisory shall be rised to the driver.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.313  | The DC-LCB shall be allowed to close when the line voltage detection DC 3kV is $\geq 1.975$ kV and $\leq 3.9$ kV (operation in Italy: $\leq 4.2$ kV, operation in Belgium: $\leq 3.8$ kV) or line voltage detection DC 1.5kV $\geq 1.0$ kV and $< 1.975$ kV AND when system isolating switch (4a) is closed!                                                                                                                                                                                                                                                                                                           | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.501  | When DC line voltage system 3.0 kV is selected and the line voltage detection DC 3kV is $\geq 1.975$ kV and $\leq 3.9$ kV (operation in Italy: $\leq 4.2$ kV, operation in Belgium: $\leq 3.8$ kV) OR DC line voltage system 1.5 kV is selected and line voltage detection DC 1.5kV is $\geq 1.0$ kV and $< 1.975$ kV, the system isolating switch (4a) shall be closed.                                                                                                                                                                                                                                               | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.314  | An event shall be set if the LCB cannot be closed due to the line voltage being out of the stated limits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Derived Requirement   | 4S_04.-.Power Supply        | EVENT          | N/A   |

|                     |                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.328                                                                                                                                                                                        |
| <b>Description</b>  | When the maximum line voltage for the nominal 25kV system is exceeded; after 10 minutes operation at line voltage above 29kV or after 1 second at line voltage above 31 kV; the line circuit breaker shall be opened. |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                 |

| Nr                                  | Description                                                                                                                                               | Applicability Details | Derived to           | User Interface | SIL   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1082 | If Line voltage AC 50 Hz is $> 32.0$ kV for $\geq 1$ sec the pantograph shall be requested to lower and an event shall be set.                            | Derived Requirement   | 4S_04.-.Power Supply | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.315  | If the AC LCB is closed and the Line voltage AC 50 Hz is $\geq 31.1$ kV for $\geq 1$ sec or $\geq 29.1$ kV for $\geq 10$ minutes the LCB shall be opened. | Derived Requirement   | 4S_04.-.Power Supply |                | SIL 2 |

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|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|-------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.578 | If the AC LCB is closed and the Line voltage AC 50 Hz is $\geq 31.1$ kV for $\geq 1$ sec or $\geq 29.1$ kV for $\geq 10$ minutes an event shall be set.                                                                                                                                                                                                            | Derived Requirement | 4S_04.-.Power Supply | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.319 | The 10 minute clock for line voltage AC 50 Hz supervision shall have following behaviour:<br><ul style="list-style-type: none"> <li>line voltage <math>\geq 29.1</math> kV, clock will be counted-up</li> <li>line voltage <math>\leq 29.0</math> kV or LCB is open, clock will be resetted</li> <li>max. time is 10 minutes, min. value for clock is 0</li> </ul> | Derived Requirement | 4S_04.-.Power Supply |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.316 | If the DC LCB is closed and the Line voltage DC 3kV $> 4.1$ kV (operation in Italy: $> 4.2$ kV, operation in Belgium: $> 4.0$ kV) for $\geq 1$ s or $> 4.0$ kV for $\geq 5$ minutes the LCB shall be opened.                                                                                                                                                       | Derived Requirement | 4S_04.-.Power Supply |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.580 | If the DC LCB is closed and the Line voltage DC 3kV $> 4.1$ kV (operation in Italy: $> 4.2$ kV, operation in Belgium: $> 4.0$ kV) for $\geq 1$ s or $> 4.0$ kV for $\geq 5$ minutes an event shall be set.                                                                                                                                                         | Derived Requirement | 4S_04.-.Power Supply | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.503 | The 5 minute clock for line voltage DC 3 kV supervision shall have following behaviour:<br><ul style="list-style-type: none"> <li>line voltage <math>&gt; 4.0</math> kV, clock will be counted-up</li> <li>line voltage <math>\leq 4.0</math> kV, clock will be counted-down</li> <li>max. time is 5 minutes, min. value for clock is 0</li> </ul>                 | Derived Requirement | 4S_04.-.Power Supply |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.317 | If the AC LCB is closed and the Line voltage AC 50Hz is $< 17.0$ kV or Line voltage AC 16.7Hz is $< 10.5$ kV for $\geq 1$ sec the LCB shall be to open.                                                                                                                                                                                                            | Derived Requirement | 4S_04.-.Power Supply |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.582 | If the AC LCB is closed and the Line voltage AC 50Hz is $< 17.0$ kV or Line voltage AC 16.7Hz is $< 10.5$ kV for $\geq 1$ sec an event shall be set.                                                                                                                                                                                                               | Derived Requirement | 4S_04.-.Power Supply | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.318 | If the DC LCB is closed and the Line voltage 3kV is $< 1.8$ kV for $\geq 1$ sec the LCB shall be opened.                                                                                                                                                                                                                                                           | Derived Requirement | 4S_04.-.Power Supply |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.583 | If the DC LCB is closed and the Line voltage 3kV is $< 1.8$ kV for $\geq 1$ sec the LCB shall be opened and an event shall be set.                                                                                                                                                                                                                                 | Derived Requirement | 4S_04.-.Power Supply | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.541 | If the DC LCB is closed and the Line voltage 1.5kV is $< 0.85$ kV for $\geq 1$ sec the LCB shall be opened.                                                                                                                                                                                                                                                        | Derived Requirement | 4S_04.-.Power Supply |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.584 | If the DC LCB is closed and the Line voltage 1.5kV is $< 0.85$ kV for $\geq 1$ sec an event shall be set.                                                                                                                                                                                                                                                          | Derived Requirement | 4S_04.-.Power Supply | EVENT | N/A   |

|                     |                                                                        |
|---------------------|------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.793                                         |
| <b>Description</b>  | The vehicle shall display the actual line voltage value to the driver. |
| <b>Safety level</b> | SIL 0                                                                  |

| Nr                                 | Description                                                                                                          | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.321 | The line voltage shall be displayed on IDU.                                                                          | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.987 | If line voltage is not detected within 5 seconds from when pantograph was indicated as raised an event shall be set. | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A   |

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### 3.3.2 Line Current Supervision

#### 3.3.2.1 Function Design

##### 3.3.2.1.1 Normal Condition

AC line current measurement is done by line current transformer (9d) placed in AC HV-box on T4 and T5 car. It is connected to the LIM which is responsible for overcurrent detection.

DC line current measurement is done by DC line current sensor (9c).

At each main transformer a current transformer for line converter control is installed.

##### 3.3.2.1.2 Failure Condition

- 1) No detection of line current when line current is expected.
- 2) No detection of transformer current when transformer current is expected.
- 3) Detection of line current when line current is not expected.

##### 3.3.2.1.3 Failure Supervision

- 1) Event will be set and driver has to change pantograph.
- 2) Depending on the situation the transformer will be isolated, bypassed or event will be set. Handled by internal propulsion supervision.
- 3) Same action as LCB does not open on order

#### 3.3.2.2 Requirements

|                     |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.460                                                             |
| <b>Description</b>  | At each main transformer there shall be a current transformer to measure the line current. |
| <b>Safety level</b> | SIL 1                                                                                      |

| Nr                                  | Description                                                                                                                                                                                                          | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.106  | The current shall be measured by a return current transformer on the primary winding at each main transformer for the differential protection and the control of the LCMs.<br>Note: Only applicable for AC operation | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.988  | The transformer currents (input and return current) shall be displayed on the IDU.                                                                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1129 | If an differential overcurrent is detected, the LCB shall be opened and reclosing shall be blocked for 5 seconds.                                                                                                    | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 2 |

|                     |                                                                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.251                                                                                                                                                                                                                  |
| <b>Description</b>  | Current measuring unit is used for line current measurement on high voltage side. The output signal shall be used for train protection (over currents, short circuits, differential currents) and for control purpose (line converter control). |
| <b>Safety level</b> | SIL 1                                                                                                                                                                                                                                           |

In AC line voltage system as well as in DC line voltage system there is a current measurement device installed in front of the LCB. The over current detection for AC is handled by the LIM and for DC by the LCB itself.

| Nr                                  | Description                                     | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|-------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1130 | The line current shall be displayed on the IDU. | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 0 |

|                                    |                         |                         |                     |                        |
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| 2F_04.01.Zefiro-Europe.CONCEPT.225 | If an overcurrent is detected, the LCB shall be opened and reclosing shall be blocked for 5 seconds.                                                                                                                                                                                                                                                                                   | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.361 | First attempt of reclosing LCB:<br>If overcurrent is detected within 5 seconds after first re-closure of the LCB, the LCB shall be opened again. The roof line disconnector related to the active pantograph shall be ordered to open.<br>After reactivation of the line voltage system the same pantograph shall be used.                                                             | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.362 | second attempt of reclosing LCB:<br>If overcurrent is detected within 5 seconds after second re-closure of the LCB, the LCB shall be opened again. The highvoltage system in the active train half shall be blocked and the other train half shall be used when next activation order is given.<br>If overcurrent is not detected on second attempt there is no further action needed. | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.363 | If an overcurrent is detected an event shall be set.                                                                                                                                                                                                                                                                                                                                   | Derived Requirement | 4S_04.-.Power Supply        | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.728 | The maximum current drawn by the vehicle in line voltage system shall be limited according the value set by the driver on IDU.                                                                                                                                                                                                                                                         | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | N/A   |

|                     |                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.773                                                                                                                    |
| <b>Description</b>  | The maximum allowed power to be extracted from the catenary by the train (also in multiple) shall be possible to be set by the driver at the IDU. |
| <b>Safety level</b> | SIL 0                                                                                                                                             |

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1050 | The driver shall be able to limit the train current by enabling a current limiter device on the IDU:<br>a) in 25 kV AC line voltage systems by steps of 20 A (in range of 20 to 560 A)<br>b) in DC 3 kV line voltage systems by steps of 100 A (in range of 100 to 3000 A)<br>c) in DC 1.5 kV line voltage systems by steps of 100 A (in range of 100 to 3200 A)<br>In case of 15 kV line voltage system: 700 A<br><br>In case of double traction the limits are doubled. | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1051 | Current shall be limited according to given values from TCMS.                                                                                                                                                                                                                                                                                                                                                                                                             | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1152 | TCMS shall set maximum level of current limitation after powered on:<br>a) in 25 kV AC line voltage systems: 560 A<br>b) in DC 3 kV line voltage system: 3000 A<br>c) in DC 1.5 kV line voltage system: 3200 A<br>In case of 15 kV line voltage system: 700 A<br><br>In case of double traction the limits are doubled.<br><br>Every time the catenary change, the max value for that catenary must be settled.                                                           | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

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### 3.3.3 Line Interference Monitor

#### 3.3.3.1 Function Design

##### 3.3.3.1.1 Normal Condition

Line interference monitor supervises interference currents (e.g. 50Hz supervision in Italian DC line), see details in TRD appendix Q.

##### 3.3.3.1.2 Failure Condition

- 1) Failure to detect line voltage.
- 2) Failure to measure overcurrent.
- 3) Other failure of LIM.

##### 3.3.3.1.3 Failure Supervision

- 1) Changeover to other HV system. Supervision by PCU.
- 2) Changeover to other HV system. Supervision by PCU.
- 3) LIM is bypasses, supervision by PCU.

#### 3.3.3.2 Requirements

|                     |                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.309                                                                                |
| <b>Description</b>  | One line interference monitor per half-train shall be installed. (e.g. 50 Hz supervision in Italian DC line). |
| <b>Safety level</b> | SIL 2                                                                                                         |

| Nr                                 | Description                                                                                                                                                                                                        | Applicability Details | Derived to           | User Interface | SIL |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.112 | At least one line interference monitor per half-train shall be installed for the supervision of the harmonic content for the DC high voltage line and for the AC high voltage line. See details in TRD appendix Q. | Derived Requirement   | 4S_04.-.Power Supply |                | N/A |

|                     |                                                                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.691                                                                                                                                                                                                                                                    |
| <b>Description</b>  | The LIM shall realize the following functions: <ul style="list-style-type: none"> <li>– Line current measurement</li> <li>– Return current measurement</li> <li>– Line voltage measurement</li> <li>– Supervision of Transformer oil temperatures</li> <li>– Self test</li> </ul> |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                                                             |

| Nr                                 | Description                                                                                                                                                                                                                   | Applicability Details | Derived to           | User Interface | SIL   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.231 | The LIM shall realize the following functions: <ul style="list-style-type: none"> <li>– Line current measurement AC</li> <li>– Return current measurement</li> <li>– Line voltage measurement</li> <li>– Self test</li> </ul> | Derived Requirement   | 4S_04.-.Power Supply |                | SIL 2 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.989  | <p>The LIM Self test shall be requested when:<br/>The Pantographs are lowered and the LCBs are open and any of the following conditions are fulfilled:</p> <ul style="list-style-type: none"> <li>– Train has started up (after closing of BC1) AND more than <b>15 hours</b> have passed since last LIM self test.</li> <li>– The self test request is set from the LIM and the train is at standstill and no cab is active.</li> <li>– The self test timeout is set from LIM. In this case the LCB shall be ordered to open and the pantograph shall be lowered in the same 8-car train.</li> <li>– <b>The cab is activated to exit of parking mode and more than 40 hours have passed since last LIM self test. In this case the LCB shall be ordered to open and the pantograph shall be lowered in the train. An event shall be set to inform the driver</b></li> </ul> <p>The self test request will be set <b>40hours</b> after last successful test. The self test timeout will be set after 50 hours and the LIM will start the test independent of TCMS.</p> <p>Info: It takes approximately 45s for the LIM to start after power on. It takes approximately 30s for the LIM to perform the test but in case of retry from LIM a time out should be min 75s.</p> | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1131 | <p>As long as the LIM self test is ongoing the LCB shall not be enabled to close and the pantograph shall not be raised.</p> <p>If the driver request to raise the pantograph while the LIM self test is ongoing an event shall be set "LIM self test ongoing" to inform the driver that the pantograph will be raised when the test is finished. The event shall be reset when LIM self test is finished or the driver request to raise the pantograph is removed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1132 | An event shall be set "LIM self test finished" when the LIM self test is finished to inform the driver that it is possible to raise the pantograph and to close the LCB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1087 | In case of LIM self test error, LIM will be bypassed and pantograph isolated (100% traction effort remaining).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1088 | In case LIM self test will not be performed although it is requested, LIM will be bypassed and pantograph isolated (100% traction effort remaining).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.756  | If an self test error is indicated, an EVENT shall be set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Derived Requirement | 4S_04.-.Power Supply        | EVENT | N/A   |

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|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.692                                                                                                                                                                                                                                                                                     |
| <b>Description</b>  | <p>The line circuit breaker shall open if</p> <ul style="list-style-type: none"> <li>– limits for line current exceeded (limits depending on country of operation)</li> <li>– Main transformer differential current exceeded</li> <li>– Short circuit detected</li> <li>– maximum line voltage exceeded</li> </ul> |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                                                                                              |

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| Nr                                 | Description                                                                                                                                                                                                                                                         | Applicability Details | Derived to           | User Interface | SIL   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.234 | The line circuit breaker shall open if <ul style="list-style-type: none"> <li>limits for line overcurrents in AC exceeded</li> <li>Main transformer differential current exceeded</li> <li>Short circuit detected</li> <li>maximum line voltage exceeded</li> </ul> | Derived Requirement   | 4S_04.-.Power Supply |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.757 | An EVENT shall be set if limits for line overcurrent exceeded.                                                                                                                                                                                                      | Derived Requirement   | 4S_04.-.Power Supply | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.758 | An EVENT shall be set when main transformer differential current exceeded.                                                                                                                                                                                          | Derived Requirement   | 4S_04.-.Power Supply | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.759 | An EVENT shall be set when short circuit detected.                                                                                                                                                                                                                  | Derived Requirement   | 4S_04.-.Power Supply | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.760 | An EVENT shall be set when maximum line voltage exceeded.                                                                                                                                                                                                           | Derived Requirement   | 4S_04.-.Power Supply | EVENT          | N/A   |

### 3.3.4 LIM 50Hz protection bypass

#### 3.3.4.1 Function Design

In case of detection of 50Hz current harmonic when the train is running on a DC line, the LIM protection system opens the line circuit breaker to avoid potential risks.

Most probably this is not due to a failure of the train propulsion system but to some incoming noise directly from the power line. The Italian regulation states that is possible to bypass this protection allowing the train to move to a different segment of line where this problem is not present.

This is allowed only when the train is running on a DC powered line.

A sealed switch is installed in the driver's cab to let the driver perform this action. The activation of this switch is managed by TCMS safe layer that communicates directly to the LIM system to bypass this protection.

##### 3.3.4.1.1 Normal Condition

In normal condition the switch is sealed and the 50Hz supervision functionality is not bypassed.

##### 3.3.4.1.2 Failure Condition

- 1) The switch is turned but no bypass has been enabled.
- 2) The switch is not turned and the bypass has been enabled.

##### 3.3.4.1.3 Failure Supervision

- 1) By the driver, if the protection occurs with the switch turned and the LCB cannot be closed.
- 2) By the driver, if the IDU indication is on without switch turned.

#### 3.3.4.2 Requirements

|                     |                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.792                                                                                     |
| <b>Description</b>  | A sealed switch to bypass the LIM 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab. |
| <b>Safety level</b> | SIL 2                                                                                                              |

| Nr                                  | Description                                                                                                        | Applicability Details | Derived to                                          | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1146 | A sealed switch to bypass the LIM 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab. | Derived Requirement   | 4S_08.07.01.-.Drivers Desk                          |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1147 | The LIM bypass switch shall be connected to safe and redundant Digital Inputs.                                     | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | SIL 2 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1148 | When the sealed switch is turned TCMS shall send the "Command: Disable LIM 50Hz supervision" to Propulsion system via MVB.                                                                                                                                   | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1149 | When the LIM bypass is activated an event shall be set.                                                                                                                                                                                                      | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1150 | When the LIM bypass is activated an indication shall be displayed on the IDU.                                                                                                                                                                                | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU   | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1151 | The 50Hz supervision of LIM (on DC line only) shall be bypassed when commanded by TCMS. This has to be done also in multiple composition. Once the by pass is activated in one cab this by pass has to be done on both vehicles of the multiple composition. | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 2 |

### 3.3.5 Neutral Section Passing

#### 3.3.5.1 Function Design

##### 3.3.5.1.1 Normal Condition

Neutral sections are powerless parts of the catenary system. There are two different types of neutral sections which are defined according to EN50388:

- AC phase separation sections, which separate between two different phases of AC line voltage system.  
In Italy, crossing AC phase separation sections will be performed automatically by indications from ATP.
- system separation sections, which separate between different line voltage systems (AC 25kV, AC 15kV, DC 3kV, DC 1.5kV)  
In Italy, crossing system separation sections will be initiated by ATP but driver has to select line voltage manually.

##### 3.3.5.1.2 Failure Condition

- AC phase separation section is not detected (ATP is faulty/isolated).
- System separation section is not detected (ATP is faulty/isolated).
- AC phase separation section detected but LCBs do not open.
- System separation section detected but LCBs do not open.
- System separation section detected but pantograph does not lower

##### 3.3.5.1.3 Failure Supervision

- Driver has to open LCB manually (Open LCB switch) before entering neutral section.
- Driver has to open LCB manually (Open LCB switch) and lower the pantograph before entering neutral section.
- Driver has to open LCB manually (Open LCB switch) before entering neutral section. When LCB is faulty, panto will lower immediately.
- Driver has to open LCB manually (Open LCB switch) before entering neutral section.  
When LCB is faulty, panto will lower immediately.
- Driver has to lower pantograph manually (Pantograph switch to "down position") after LCB opened but before entering neutral section.

#### 3.3.5.2 Requirements

|                     |                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.463                                                                                                           |
| <b>Description</b>  | The AC phase separation sections and separation section sequences shall be activated by External sensors e.g. ATP or track side magnets. |
| <b>Safety level</b> | SIL 0                                                                                                                                    |

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| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.464                                                                                                                                                                                  |
| <b>Description</b>  | When the AC phase separation section sequence is activated, the Line Converters shall be ramped down and the line circuit breakers shall be opened for the time of passage of the AC phase separation sections. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                           |

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Applicability Details | Derived to                 | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.515  | AC phase separation section sequence shall be activated by external sensors (e.g. ATP).                                                                                                                                                                                                                                                                                                                                                                          | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1112 | System separation section sequence shall be activated by external sensors (e.g. ATP).                                                                                                                                                                                                                                                                                                                                                                            | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1122 | In Italy, the sequence of crossing AC phase separation sections shall start when external sensors orders "traction block" and "LCB opening".                                                                                                                                                                                                                                                                                                                     | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1123 | In Italy, the sequence of crossing system separation sections shall start when external sensors orders "traction block", "LCB opening" and "pantograph down".                                                                                                                                                                                                                                                                                                    | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.521  | For country options, the sequence of crossing AC phase separation sections or system separation sections shall start when the train has travelled the distance between the sensor in the leading unit and the raised pantograph.                                                                                                                                                                                                                                 | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.510  | The power reference shall be ramped down to 0 before requesting the LCB to open prior to enter AC phase separation section or system separation section.                                                                                                                                                                                                                                                                                                         | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.441  | The LCB shall be requested to be opened before entering the AC phase separation section or system separation section.                                                                                                                                                                                                                                                                                                                                            | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.442  | When the LCB has opened for AC phase separation section or system separation section it shall stay open for at least 4 seconds.                                                                                                                                                                                                                                                                                                                                  | Derived Requirement   | 4S_04.-Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.512  | AC phase separation section or system separation section sequence shall be reset if the train is standing still.                                                                                                                                                                                                                                                                                                                                                 | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.513  | The AC phase separation section or system separation section sequence shall be reset on the return of line voltage (positive edge)                                                                                                                                                                                                                                                                                                                               | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.344  | If LCB was opened on order from external sensors (ATP) before entering the AC phase separation section, the LCB shall be requested to be closed again after passing the AC phase separation section, automatically when ordered by external sensors only if any command to open hasn't been given from driver or internal protection of HV equipment and propulsion system, and system detection has confirmed selected line voltage system (see chapter 3.4.1). | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.776  | If LCB was opened on order from external sensors (ATP) before entering the system separation section, the LCB shall be requested to be closed again after passing the system separation section, automatically when ordered by external sensors only if any command to open hasn't been given from driver or internal protection of HV equipment and propulsion system, and system detection has confirmed selected line voltage system (see chapter 3.4.1).     | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.443  | After the LCB has been closed automatically when ordered by external sensors (ATP), MCM power reference shall be ramped up.                                                                                                                                                                                                                                                                                                                                      | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | N/A   |

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| 2F_04.01.Zefiro-Europe.CONCEPT.196 | If the train is in single or multiple operation under 3 kV and the ATP is communication a voltage change (the catenary that will come is different than 3 kV) and if the voltage is going to 0 and the panto of the leading train is up. TCMS must command a fast panto down to all the panto in the composition.<br>This to avoid that the train in sigle or multiple composition will run with the 3 kV panto raised under 25 kV. | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | N/A |
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Passing a system separation section will be done in the same way as a AC phase separation section with the extra function to swap pantograph before reconnection on a different line voltage system.

| Nr                                 | Description                                                                                                                                    | Applicability Details | Derived to                  | User Interface | SIL |
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| 2F_04.01.Zefiro-Europe.CONCEPT.505 | If system separation section is activated, from external sensors (ATP), LCB shall be ordered to open and pantograph shall be ordered to lower. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |

The only action which is needed from driver during crossing of system separation section is the manual selection of the upcoming line voltage system (line voltage selector switch).

ATP will indicate to the driver which line voltage system will follow after the separation section.

|                     |                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.708                                                                                                                                                                                               |
| <b>Description</b>  | The AC phase separation or separation section shall be activated via External sensors (e.g. ATP). The External sensors also gives additional information to lower pantograph after the Line circuit breaker has been opened. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                        |

The Italian ATP system reports: Information about the line voltage system on the other side of the neutral section.

| Nr                                 | Description                                                                                                                                                                                                                                                                               | Applicability Details | Derived to                  | User Interface | SIL   |
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| 2F_04.01.Zefiro-Europe.CONCEPT.458 | If external sensors (ATP) indicate a AC phase separation section requires lowering of pantographs, all pantographs shall be lowered before entering the AC phase separation section. (Note this is not the same as a system separation section)                                           | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.573 | If external sensors (ATP) indicate that the voltage system after passing the AC phase / system separation section is different than the voltage system before the AC phase / system separation section, the passage shall be done according to requirement for system separation section. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

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| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.335                                                                                                                                                 |
| <b>Description</b>  | Where AC phase separation sections require pantographs to be lowered and subsequently raised, these additional actions are permitted to be automatically initiated by the ATP. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                          |

External sensors (ATP) might in some situation be able to not only order pantograph down but also pantograph up.

| Nr                                 | Description                                                                                                                                                               | Applicability Details | Derived to                  | User Interface | SIL   |
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| 2F_04.01.Zefiro-Europe.CONCEPT.523 | If external sensors (ATP) require the pantograph to be raised, the selected pantograph shall raise. (This shall work in direct parallel to the "raise pantograph switch") | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1089 | If the pantographs have been lowered on order from external sensor before entering a AC phase / system separation section and driver has not moved panto-switch to "down" position, raise-order for pantograph shall be blocked as long as the train has not passed neutral section.                                                        | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1126 | Pantograph shall only be allowed to raise, when TCMS has confirmed that selected line voltage system is in line with system which is indicated by ATP.                                                                                                                                                                                      | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1127 | It shall be indicated on IDU when selection of line voltage is not in line with ATP notification.                                                                                                                                                                                                                                           | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.462  | If the pantographs have been lowered on order from external sensor before entering a AC phase / system separation section and driver has not moved panto-switch to "down" position, raise-order for pantograph after crossing the neutral section shall be set automatically by external sensors depending on selected line voltage system. | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |

|                     |                                                                                                                                                                       |
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| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.687                                                                                                                                        |
| <b>Description</b>  | When multiple pantographs are lifted and spacing allowed in coming neutral section is exceeded, the pantographs shall be lowered before entering the neutral section. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                 |

If two or more pantographs are lifted, a neutral section with a length about the same as the distance between two pantographs can be short circuit by the carbon strip connecting the neutral catenary with the live catenary.

| Nr                                 | Description                                                                                      | Applicability Details | Derived to                  | User Interface | SIL   |
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| 2F_04.01.Zefiro-Europe.CONCEPT.518 | In multiple operation, pantographs shall raise when external sensors order "pantograph raising". | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

### 3.3.6 System Selection

#### 3.3.6.1 Function Design

##### 3.3.6.1.1 Normal Condition

The driver selects the line voltage system (AC 25kV, AC 15kV, DC 3kV or DC1.5kV) by using 4-position switch "Line voltage selector".

When line voltage system changes from one to another of these four voltage systems a system selection has to be performed. To identify exactly the country of operation, the country code needs to be selected.

The system detection units have to detect, to which line voltage system the train is connected.

##### 3.3.6.1.2 Failure Condition

- 1) System detected by AC system detection unit and LIM differ
- 2) Faulty System detected by AC system detection unit
- 3) System detected by DC system detection unit and LIM differ
- 4) Faulty System detected by DC system detection unit
- 5) Plausibility fault DC and 3kV from LIM

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### 3.3.6.1.3 Failure Supervision

- 1) Faulty system detection. Lower and isolate AC pantograph. Driver has to raise other AC pantograph,
- 2) Faulty system detection. Lower and isolate AC pantograph. Driver has to raise other AC pantograph,
- 3) Faulty system detection. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,
- 4) Faulty system detection. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,
- 5) Faulty signals from LIM. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,

### 3.3.6.2 Requirements

|                     |                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.771                                                                                                                                       |
| <b>Description</b>  | The following power supply system shall be possible to be selected by the driver at driver's desk:<br>AC 25 kV 50 Hz,<br>AC 15 kV 16.7 Hz,<br>3 kV DC,<br>1.5 kV DC. |
| <b>Safety level</b> | N/A                                                                                                                                                                  |

| Nr                                  | Description                                                                                                                                                                                                                                | Applicability Details | Derived to                                          | User Interface | SIL |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.492  | The driver shall be able to select the voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) prior to raising the pantograph.                                                                                                              | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.976  | A 4-position switch "Line voltage selector" shall be installed at the driver's desk. With this switch it shall be possible to select the following power supply systems:<br>AC 25 kV 50 Hz,<br>AC 15 kV 16.7 Hz,<br>3 kV DC,<br>1.5 kV DC. | Derived Requirement   | 4S_08.07.01.-.Drivers Desk                          |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.977  | The following positions of the "Line voltage selector" switch shall be connected to TCMS<br>AC 25 kV 50 Hz,<br>AC 15 kV 16.7 Hz,<br>3 kV DC,<br>1.5 kV DC.                                                                                 | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1069 | The voltage selection has to be stable for 1 second before TCMS uses the input signal (only applicable for AC operation).                                                                                                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A |

In addition country code is needed to identify exactly, in which country the train operates.

| Nr                                 | Description                                                                                                                                                                                                                             | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.731 | If the train is not designed for operation in all 4 line voltage systems and the driver turns "line voltage selector" to a system which is not implemented, a failure message shall be set and raise of pantographs shall be blocked.   | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.732 | If the driver turns "line voltage selector" to change the selected line voltage system without opening the LCB and lowering the pantograph before, all LCB shall be requested to open and all pantographs shall be lowered immediately. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

|                     |                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.700                                                                                          |
| <b>Description</b>  | It shall be ensured that the DC measurement device is not damaged if a DC panto is raised at an AC catenary by mistake. |
| <b>Safety level</b> | SIL 1                                                                                                                   |

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There will be a system detection device at each TT2, T4, T5 and TT7 car. Each will be able to detect as well AC and DC line voltage regardless of what pantograph it is connected to.

| Nr                                 | Description                                                                    | Applicability Details | Derived to           | User Interface | SIL   |
|------------------------------------|--------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.553 | System detection device shall detect and report both AC and DC voltage system. | Derived Requirement   | 4S_04.-.Power Supply |                | SIL 2 |

|                     |                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.337                                                                                                                                                                                                                |
| <b>Description</b>  | When pantographs are not lowered from the catenary line when passing a neutral section, only those electric circuits on the traction units, which instantaneously conform to the power supply system at the pantograph, may remain connected. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                                         |

|                     |                                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.707                                                       |
| <b>Description</b>  | The high voltage system shall protect itself against false voltage system selection. |
| <b>Safety level</b> | SIL 0                                                                                |

When the LCBs and system isolation switches are open, the only equipment that is connected to the line voltage via the pantograph is the system detection device. This device detects if the present line voltage system (AC 25kV, AC 15kV, DC 3kV, DC 1.5kV) is in line with the driver-selected line voltage system.

If present line voltage system does not comply to selected system, pantograph will be lowered and driver gets an indication to select another line voltage system.

| Nr                                 | Description                                                                                                                                                                                                                      | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.739 | In case the detected system does not fit the selected system <ul style="list-style-type: none"> <li>there shall be a message on the IDU</li> <li>the raise signal shall be canceled and the pantograph become lowered</li> </ul> | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.524 | If line voltage system is AC and 50Hz is detected propulsion system shall be set up for 25kV 50Hz.                                                                                                                               | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.525 | If line voltage system is AC and 16 2/3Hz is detected propulsion system shall be set up for 15kV 16 2/3Hz.                                                                                                                       | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.526 | If line voltage system is DC and line voltage is in the range $\geq 1.975$ kV propulsion system shall be set up for 3kV DC.                                                                                                      | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.527 | If line voltage system is DC and line voltage is in the range $\geq 0.8$ kV and $< 1.975$ kV propulsion system shall be set up for 1.5kV DC.                                                                                     | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |

|                     |                                                                                |
|---------------------|--------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.287                                                 |
| <b>Description</b>  | For AC and for DC it shall be detected which line voltage system is connected. |
| <b>Safety level</b> | N/A                                                                            |

System selection is done automatically by comparing the measured voltage system with the driver-selected voltage system. The system validates the selected voltage system via measurement before allowing the LCBs to be closed.

The system detection detects the voltage of contact wire voltage (15kV AC, 25kV AC, 3kV DC or 1.5kV DC).

| Nr                                 | Description                                                                                                                                                       | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.219 | The LCB shall be opened resceptively not closed if the system detection detects another line voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) than selected. | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.312 | An event shall be set if the line voltage system detected by Propulsion differs from the selected line voltage system by the line voltage selector.               | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A   |

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| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.702                                                                              |
| <b>Description</b>  | It shall be ensured that the main trafo is not damaged if a AC panto is raised at a DC catenary by mistake. |
| <b>Safety level</b> | SIL 1                                                                                                       |

The system detection device will prevent the LCB from closing if the measured system is not the valid voltage system for that LCB.

| Nr                                 | Description                                                                                                                                       | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.459 | It shall not be possible to close the LCB when wrong system is detected compared to the raised pantograph.                                        | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.785 | If line voltage system sent from Propulsion does not comply to selected system by the line voltage selector the closing of LCB shall be disabled. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

### 3.4 Line Circuit Breakers and system isolation switches

#### 3.4.1 Line Circuit Breaker Closing

##### 3.4.1.1 Function Design

##### 3.4.1.1.1 Normal Condition

Connecting of line voltage by closing LCBs.

##### 3.4.1.1.2 Failure Condition

- 1) LCB does not close on order.
- 2) LCB closes without order.

##### 3.4.1.1.3 Failure Supervision

- 1) If the feedback "LCB closed" is not given latest 1s after the "LCB close" order is set, the LCB is considered as faulty. The LCB will be blocked but "activation of line voltage" will retry the closing order.  
After 3 attempts without success within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.
- 2) The related pantograph will be lowered, the pantograph disconnecter will open and both components will be blocked!

##### 3.4.1.2 Requirements

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.468                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description</b>  | <p>Line circuit breaker shall be closed when</p> <ul style="list-style-type: none"> <li>– 'Line circuit breaker close' command given by driver OR automatically when leaving a neutral section AND</li> <li>– pantograph is already lifted AND</li> <li>– no interlocking for maintenance activities active AND</li> <li>– no line circuit breaker opening command</li> </ul> <p>The 'Line circuit breaker close' command is done by turning the Line Circuit Breaker switch on driver's desk to position ON.</p> |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                      | Applicability Details | Derived to                                          | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.978  | There shall be 3-position switch for closing/opening of the Line Circuit Breaker. The positions shall be:<br>1: LCB close (unstable)<br>2: neutral (stable)<br>3: LCB open (unstable)                                                                                                                                                                            | Derived Requirement   | 4S_08.07.01.-.Drivers Desk                          |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.979  | The following positions of the Line Circuit Breaker switch shall be connected to TCMS:<br>1: LCB close<br>2: LCB open                                                                                                                                                                                                                                            | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.127  | The status of the LCBs shall be displayed on the IDU:<br>– open/closed<br>– faulty (automatic block)<br>– manually blocked                                                                                                                                                                                                                                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         | IDU            | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.970  | The LCB shall be inhibited to close if there is no valid country code (country code = 0).                                                                                                                                                                                                                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.374  | When the driver turns LCB-switch to "close" position AND the pantograph is raised AND line voltage is detected for the selected line voltage system (AC 25kV, AC 15kV, DC 3kV or DC 1.5kV), the LCB which is related to the raised pantograph shall be enabled to close. LCB can be closed from driver's desk only if the master controller is on zero position. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1224 | If the LCB is requested (from driver) to close while the Master controller is out of the 0 position, an advisory shall be shown to the driver asking 'to set the master controller in 0 position before ordering LCB closure'                                                                                                                                    | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1090 | AC LCB shall only be allowed to close when both AC roof line disconnectors are in end position. In case of plausibility fault of remote AC roof line disconnector, AC LCB shall only be allowed to close when the local disconnector is in open position.                                                                                                        | Derived Requirement   | 4S_04.-.Power Supply                                |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1091 | DC LCB shall only be allowed to close when both DC bus bar disconnectors are in end position. In case of plausibility fault of DC roof line disconnector the train behaviour has to be the following:<br>– in case the fault happens in operation, no action<br>– in case the fault happens at train startup, both roof disconnectors have to be ordered open    | Derived Requirement   | 4S_04.-.Power Supply                                |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.992  | The LCB shall only be allowed to be closed when the LCB enable signal is set from TCMS.                                                                                                                                                                                                                                                                          | Derived Requirement   | 4S_04.-.Power Supply                                |                | N/A   |

There are situations when the LCB will remain open even though TCMS is enabling closing of the LCB. The decision to close the LCB will be based on internal Propulsion supervision and the enable order from TCMS.

| Nr                                 | Description                                                                                                           | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.500 | The system isolation switch shall be closed before the DC LCB can be closed.                                          | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.381 | When ever the "LCB Close" order is removed, there shall be a 4 second delay before the LCB is allowed to close again. | Derived Requirement   | 4S_04.-.Power Supply        |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.386 | The number of closings of each LCB shall be stored as condition data.                                                 | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

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|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------|-------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.532  | When the "LCB closed" feedback and "LCB not closed" feedback indicates different states (one indicates closed and one open) for more than 1 second the LCB shall be ordered to open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Derived Requirement | 4S_04.-.Power Supply                                |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.387  | When the "LCB closed" feedback and "LCB not closed" feedback indicates different states (one indicates closed and one open) for more than 1 second an event shall be set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Derived Requirement | 4S_04.-.Power Supply                                | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.531  | When the "LCB closed" feedback and "LCB not closed" feedback indicates different states (one indicates closed and one open) for more than 1 second IDU shall show a faulty LCB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Derived Requirement | 4S_09.03.05.-.TCMS Software                         | IDU   | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.761  | When the feedback "LCB closed" is not given latest 1 second after the "LCB close" order is set, the LCB is considered as faulty. The LCB shall be blocked, but operation of "Close LCB" switch shall retry the closing order.<br>After 3 attempts without success within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Derived Requirement | 4S_04.-.Power Supply                                |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.762  | If the LCB is closed without order, the related pantograph shall be lowered, the pantograph disconnector shall open and both components shall be blocked!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1161 | After manual exit from parking mode, LCB under selected pantograph has to be closed and LCB under not selected pantograph has to be opened.<br>– In case the new configuration is different from the LCB closed during parking mode and DC line voltage system is selected, the LCB under selected pantograph shall be automatically closed and afterwards the LCB under not selected pantograph shall open.<br>In case in the new configuration the driver has requested raising of both pantographs, this procedure lets closed the LCB which is related to the rear pantograph.<br>– In case the new configuration is different from the LCB closed during parking mode and AC line voltage system is selected, LCB closing command will not be given automatically by TCMS (Driver has to do it manually)<br>This procedure can be only performed at standstill. If the train starts running, this procedure will be stopped immediately opening both LCBs. | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1162 | While manual exit from parking mode procedure is active, LCB shall close on request from TCMS, if all the other conditions are fulfilled, and it allowed to close both LCBs for a short time only (< 2 s).<br>The LCB which is no longer requested to close from TCMS, shall open without any interruption of the auxiliary power supply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Derived Requirement | 4S_04.-.Power Supply                                |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1164 | The voltage coherence of the two pantographs shall be confirmed, before to allow both LCB to be closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.388  | In operation mode (refer to key interlocking system), closing of LCB shall be only possible if pantograph is raised and selected line voltage system is confirmed by system detection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Derived Requirement | 4S_09.03.05.-.TCMS Software                         |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.495  | In maintenance mode (refer to key interlocking system), closing of LCB shall be possible if ALL pantographs are lowered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits |       | SIL 2 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1177 | An event shall be set in case the maintenance mode relays have discrepancy between DM1 and DM8 on the same trainset                                                                                                                                                             | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1059 | In maintenance mode (refer to key interlocking system), closing of LCB shall be possible if ALL pantographs are lowered.                                                                                                                                                        | Derived Requirement | 4S_04.-.Power Supply        |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.796  | LCB shall only be allowed to close when converter is disconnected or DC-link is charged to avoid hard switch-on of converters.                                                                                                                                                  | Derived Requirement | 4S_04.-.Power Supply        |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.389  | When external power supply is active the LCB shall not be allowed to close.                                                                                                                                                                                                     | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1214 | If the driver, with train standstill, request any panto to be rised a warning to the driver shall be set. The warning alerts the driver that at least one 400V external plug is inserted and in case of false reporting the plug can be manually isolated using softkey on iDU. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |

|                     |                                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.703                                                        |
| <b>Description</b>  | Line circuit breaker shall only be allowed to close if valid line voltage is detected |
| <b>Safety level</b> | SIL 1                                                                                 |

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.347  | Line circuit breaker shall only be allowed to close if valid line voltage is detected or train is in maintenance mode.<br>See section 3.4.1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Derived Requirement   | 4S_04.-.Power Supply        |                | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1163 | During parking mode, in case of missing line voltage the automatic exit procedure starts. In case the line voltage is detected again, the automatic exit procedure shall be stopped and LCB shall be re-closed (only LCB already closed before missing line voltage, no other LCB shall be closed)<br>To avoid simultaneous closing of many trains, LCB shall be automatically closed after a defined amount of time from the line voltage return that is proportional to the train number: time delay in seconds = (train number) + 5, where (train number) is integer in the range 0 to 99, considering the two least significant digits only. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

|                     |                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.623                                                                     |
| <b>Description</b>  | The line circuit breakers of the train shall be closed one after another to limit inrush currents. |
| <b>Safety level</b> | SIL 0                                                                                              |

| Nr                                 | Description                                                                                            | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.249 | If there are 2 trains in multiple the "LCB Close" order shall be delayed 1 second in the second train. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

|                     |                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.705                                                                                                                                               |
| <b>Description</b>  | The Line circuit breaker closure shall be synchronised to close at the top of the sinusoidal line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo. |
| <b>Safety level</b> | SIL 1                                                                                                                                                                        |

| Nr                                 | Description                                                                                                                                                                  | Applicability Details | Derived to           | User Interface | SIL |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.350 | The Line circuit breaker closure shall be synchronised to close at the top of the sinusoidal line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo. | Derived Requirement   | 4S_04.-.Power Supply |                | N/A |

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## 3.4.2 Line Circuit Breaker Opening

### 3.4.2.1 Function Design

#### 3.4.2.1.1 Normal Condition

Disconnection of Line Voltage will be done by LCB.

#### 3.4.2.1.2 Failure Condition

- 1) LCB does not open on order.
- 2) LCB opens without order.

#### 3.4.2.1.3 Failure Supervision

- 1) The related pantograph will be lowered, the pantograph disconnecter will open and both components will be blocked!
- 2) If the LCB opens without order it will be ordered to close again. If the failure is present 3 times within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.

### 3.4.2.2 Requirements

|                     |                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.234                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Description</b>  | Line circuit breaker shall be opened when <ul style="list-style-type: none"> <li>– open command from LIM is received OR</li> <li>– line voltage system detection protection or</li> <li>– interlocking for maintenance activities is active OR</li> <li>– drivers emergency push button was pressed OR</li> <li>– ADD on pantograph or</li> <li>– protective shut down from any DCU.</li> </ul> |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                                                                                                                                                                           |

| Nr                                  | Description                                                                                                                                                           | Applicability Details | Derived to                                              | User Interface | SIL   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.408  | If a line trip is detected the LCB shall be opened.                                                                                                                   | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.750  | If a power supply / propulsion line trip is detected the LCB shall be opened.                                                                                         | Derived Requirement   | 4S_04.-.Power Supply                                    |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.409  | If the "Emergency push button" is pushed the LCB shall be opened via Train Line ("LCB Emergency Off").                                                                | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.751  | If the "Emergency push button" is pushed the LCB shall be opened via Train Line ("LCB Emergency Off").                                                                | Derived Requirement   | 4S_04.-.Power Supply                                    |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1179 | If the "Emergency push button" is pushed and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set         | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1071 | The LCB shall open immediately, when pantograph down order is given.                                                                                                  | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.410  | The LCB shall open automatically, if any A-Key is activated (key interlocking system).                                                                                | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.752  | The LCB shall open automatically, if any A-Key is activated (key interlocking system).                                                                                | Derived Requirement   | 4S_04.-.Power Supply                                    |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1180 | If any A-Key is activated and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1181 | When IDU soft key is pressed from maintainer, a test of LCB Emergency Off trainline activation shall be carried out (energizing dedicated DOs only in the active cab) | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             | IDU            | N/A   |

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|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------|-------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1182 | Automatically every time when the LIM test is performed, a test of LCB Emergency Off trainline activation shall be carried out (energizing dedicated DOs only in one cab at the head of composition) | Derived Requirement | 4S_09.03.05.-TCMS Software |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1183 | If LCB Emergency Off trainline activation test is performing and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set                    | Derived Requirement | 4S_09.03.05.-TCMS Software | EVENT | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1092 | When one AC roof line disconnecter leaves end position, AC LCB shall be ordered to open.                                                                                                             | Derived Requirement | 4S_04.-.Power Supply       |       | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1093 | When one DC bus bar disconnecter leaves end position an event shall be set when plausibility fault is detected.                                                                                      | Derived Requirement | 4S_04.-.Power Supply       |       | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.763  | When LCB does not open on order, the related pantograph shall be lowered, the pantograph disconnecter shall open and both components shall be blocked!                                               | Derived Requirement | 4S_09.03.05.-TCMS Software |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.764  | When the LCB opens without order, it shall be ordered to close again. If the failure is present 3 times within 30 minutes the LCB will be blocked permanently and the pantograph shall be blocked.   | Derived Requirement | 4S_04.-.Power Supply       |       | N/A   |

|                     |                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.704                                                                                                                                                              |
| <b>Description</b>  | Line circuit breaker shall be opened when<br><ul style="list-style-type: none"> <li>– 'Line circuit breaker open' command given by driver OR</li> <li>– entering neutral section</li> </ul> |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                       |

| Nr                                  | Description                                                                                                                                                                                                        | Applicability Details | Derived to                 | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.353  | When the Line Circuit Breaker switch is turned to "LCB open" the LCB shall be ordered to open (also in rear train in case of multiple operation).                                                                  | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1056 | LCB shall open when country code becomes invalid (country code = 0).                                                                                                                                               | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.403  | A permanently blocked (due to a failure) LCB shall be blocked until a master reset is made or PCU is restarted.                                                                                                    | Derived Requirement   | 4S_04.-.Power Supply       |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.995  | If LCB is permanently blocked (due to failure), related pantograph shall be ordered to lower and related pantograph disconnecter shall open and both, pantograph and disconnecter, shall be blocked.               | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.406  | If the feedback 'LCB Closed' changes to 'LCB not closed' while the 'LCB close' order is high, the status of the LCB is inconsistent and the LCB shall be ordered to open and the LCB shall be permanently blocked. | Derived Requirement   | 4S_04.-.Power Supply       |                | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.537  | If the feedback 'LCB Closed' changes to 'LCB not closed' while the 'LCB Close' order is high, the status of the LCB is inconsistent and an event shall be set.                                                     | Derived Requirement   | 4S_04.-.Power Supply       | EVENT          | N/A   |

|                                    |                         |                         |                     |                        |
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| 2F_04.01.Zefiro-Europe.CONCEPT.986 | <p>If all LCBs in the trains are open (valid also for Multiple Unit), the 'LCB is open' indication lamp shall be illuminated on driver's desk.</p> <p>For more details see the tables below for multiple unit and single unit:</p> <p>single unit:</p> <table><tr><th colspan="3">SINGOLO</th></tr><tr><th>IR1</th><th>IR2</th><th>spia</th></tr><tr><td>0</td><td>0</td><td>accesa</td></tr><tr><td>0</td><td>1</td><td>spenta</td></tr><tr><td>1</td><td>0</td><td>spenta</td></tr><tr><td>1</td><td>1</td><td>spenta</td></tr></table> <p>multiple unit:</p> <table><tr><th>IR1</th><th>IR2</th><th>IR3</th><th>IR4</th><th>spia</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>accesa</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>LAMP</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>LAMP</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>LAMP</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>LAMP</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>spenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>spenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>spenta</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>LAMP</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>spenta</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>spenta</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>spenta</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>LAMP</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>spenta</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>spenta</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>spenta</td></tr></table> <p>0 = aperto<br/>1 = chiuso</p> | SINGOLO             |                                                    |        | IR1   | IR2 | spia | 0 | 0 | accesa | 0 | 1 | spenta | 1 | 0 | spenta | 1 | 1 | spenta | IR1 | IR2 | IR3 | IR4 | spia | 0 | 0 | 0 | 0 | accesa | 0 | 0 | 0 | 1 | LAMP | 0 | 0 | 1 | 0 | LAMP | 0 | 0 | 1 | 1 | LAMP | 0 | 1 | 0 | 0 | LAMP | 0 | 1 | 0 | 1 | spenta | 0 | 1 | 1 | 0 | spenta | 0 | 1 | 1 | 1 | spenta | 1 | 0 | 0 | 0 | LAMP | 1 | 0 | 0 | 1 | spenta | 1 | 0 | 1 | 0 | spenta | 1 | 0 | 1 | 1 | spenta | 1 | 1 | 0 | 0 | LAMP | 1 | 1 | 0 | 1 | spenta | 1 | 1 | 1 | 0 | spenta | 1 | 1 | 1 | 1 | spenta | Derived Requirement | 4S_09.03.05.-TCMS Software |  | SIL 0 |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------|--------|-------|-----|------|---|---|--------|---|---|--------|---|---|--------|---|---|--------|-----|-----|-----|-----|------|---|---|---|---|--------|---|---|---|---|------|---|---|---|---|------|---|---|---|---|------|---|---|---|---|------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---------------------|----------------------------|--|-------|
| SINGOLO                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                                                    |        |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| IR1                                | IR2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | spia                |                                                    |        |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | accesa              |                                                    |        |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | spenta              |                                                    |        |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | spenta              |                                                    |        |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | spenta              |                                                    |        |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| IR1                                | IR2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | IR3                 | IR4                                                | spia   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 0                                                  | accesa |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 1                                                  | LAMP   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 0                                                  | LAMP   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 1                                                  | LAMP   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 0                                                  | LAMP   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 1                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 0                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 0                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 1                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 0                                                  | LAMP   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 1                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 0                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 1                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 0                                                  | LAMP   |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                   | 1                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 0                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 1                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                   | 1                                                  | spenta |       |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.985 | When ordered by TCMS, the indication lamp 'LCB is open' shall be illuminated on the driver's desk.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Derived Requirement | 4S_09.02.09.-Conventional Train Control - Circuits | HW HMI | SIL 0 |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |
| 2F_04.01.Zefiro-Europe.CONCEPT.727 | If the cab is deactivated without entering the parking mode, all LCB shall be requested to open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Derived Requirement | 4S_09.03.05.-TCMS Software                         |        | SIL 0 |     |      |   |   |        |   |   |        |   |   |        |   |   |        |     |     |     |     |      |   |   |   |   |        |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |      |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |                     |                            |  |       |

|                     |                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.712                                                                                                            |
| <b>Description</b>  | Line circuit breaker shall be opened fast enough when system change section is entered under speed to avoid damage of electric. equipment |
| <b>Safety level</b> | SIL 1                                                                                                                                     |

If the train is passing a system separation section with a manual indication in proper time or a correct signal from external sensors, the LCB will open before entering the system separation section. See section 3.3.4

| Nr                                  | Description                                                                                                                                                                                           | Applicability Details | Derived to                                          | User Interface | SIL |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.464  | If line voltage lower than 1.8kV in DC 3kV or 0.85kV in DC 1.5kV for more than 1 second when the train speed is >30km/h the pantograph shall be requested to lower directly after the LCB has opened. | Derived Requirement   | 4S_04.-.Power Supply                                |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1105 | The pantograph shall be lowered using the pantograph fast down trainline when requested from PCU (propulsion control unit).                                                                           | Derived Requirement   | 4S_09.03.05.-TCMS Software                          |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1070 | The pantograph fast down trainline shall be set when ordered by TCMS DO.                                                                                                                              | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A |

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|---------------------|-------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.714                                                |
| <b>Description</b>  | Line circuit breaker shall open in case of risk for overtemperature for trafo |
| <b>Safety level</b> | SIL 1                                                                         |

Pre-Condition: Temperature measurement has to be installed at main transformer.

| Nr                                 | Description                                                                                                                                                                                                                                  | Applicability Details | Derived to           | User Interface | SIL   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.765 | If an over temperature in main transformer at AC operation or line inductor at DC operation is detected, the related LCB shall open and an EVENT shall be set.                                                                               | Derived Requirement   | 4S_04.-.Power Supply | EVENT          | SIL 1 |
| 2F_04.01.Zefiro-Europe.CONCEPT.497 | If an over temperature in the transformer is detected running AC for 2 times in 12 hours the high voltage section (with the over heated transformer) shall be isolated in AC operation (refer to section 3.4.3 "Isolate high voltage").      | Derived Requirement   | 4S_04.-.Power Supply |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.498 | If an over temperature in line inductor is detected running DC for 2 times time in 12 hours the high voltage section (with the over heated line inductor) shall be isolated in DC operation (refer to section 3.4.3 "Isolate high voltage"). | Derived Requirement   | 4S_04.-.Power Supply |                | N/A   |

|                     |                                                                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.624                                                                                                                       |
| <b>Description</b>  | In case of normal opening (no protection situation) the Line Converter Modules shall be ordered to ramp down before opening the line circuit breaker |
| <b>Safety level</b> | SIL 0                                                                                                                                                |

| Nr                                 | Description                                                                                          | Applicability Details | Derived to           | User Interface | SIL |
|------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.252 | Any order of normal opening of LCB requires a ramp down of line converters before LCB can be opened. | Derived Requirement   | 4S_04.-.Power Supply |                | N/A |

|                     |                                                                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.326                                                                                                                                                                                                                                             |
| <b>Description</b>  | In case of fire, the power supply to the affected system shall be interrupted. If one traction circuit is concerned, the respective line circuit breaker shall open, the affected system isolated from the traction circuit and the line circuit breaker then close again. |
| <b>Safety level</b> | SIL 0                                                                                                                                                                                                                                                                      |

| Nr                                  | Description                                                                                                                                                                                                                                                                       | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.255  | If fire is detected in the underframe the LCB shall be requested to open and the disconnectors shall be requested to open.                                                                                                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.465  | If the LCB and roof line disconnectors has opened due to fire, this half of the train where the fire is detected shall be isolated (disable closing of LCB and block raising of pantograph) and an event shall be set to inform the driver that change of pantograph is required. | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1100 | It shall be possible to manually cut out a LCB.                                                                                                                                                                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1101 | If the LCB is manually cut out the LCB an event shall be set as long as the LCB is cut out.                                                                                                                                                                                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1102 | A LCB that is manually cut out shall be opened and not closed.                                                                                                                                                                                                                    | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A   |

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| 2F_04.01.Zefiro-<br>Europe.CONCEPT.1245 | <p>After 1 sec. that any AC LCB is open and not closed in the consist (single or multiple operation) and a voltage of AC is detected, an even shall be set and the related panto must be blocked.<br/>The line voltage to take in consideration shall be the one after the AC LCB that is provided with the variable. The supervision shall be deactivated if AC LCB is closed and not open or after 10 seconds that the AC LCB is open and not closed.</p> <table><tr><th>PCU &gt;<br/>CCU-0</th><th>Signal</th><th>Cycle<br/>time</th><th>Description</th><th>Scale</th></tr><tr><td>CX1CT807</td><td>HCV_XU_AcLn</td><td>64 ms</td><td>LCM AC line voltage to remote system</td><td>1=10V</td></tr><tr><td>CX1CT8027</td><td>HCV_XU_AcLn</td><td>64 ms</td><td>Valid bit LCM AC line voltage to remote system</td><td>L</td></tr></table> <p>The function is activated if the voltage is higher than 15 kV for more than 0.5 sec ( under 25 kv operation ) after 1 sec. even if any LCB is in open position, the panto related to the LCB that was in closed position before must be put in fault.<br/>An event to the driver shall be set with the following sentence" LCB X non aperto, il pantografo e inibito e non deve essere effettuato nessun master reset". Also the buzzer shall be activated.<br/>This logic must be done on consist level, this means that if an AC LCB on the rear/front train is not opened , the message must be send to the driver in the front train where the cab is active.<br/>The message for the diver are:<br/>For the fault of the LCB in T4<br/>"Isolare pneumaticamente l'impianto di sollevamento pantografo 25 kV tramite rubinetto nel quadro 59.C4 (rubinetto 08.18/02) della carrozza segnalata"</p> <p>For the fault of the LCB in T5<br/>"Isolare pneumaticamente l'impianto di sollevamento pantografo 25 kV tramite rubinetto nel quadro 59.C4 (rubinetto 08.18/02) della carrozza segnalata"</p> <p>The eviroment data related to the event shall be the following, with a sample time of 100 ms if possible</p> <table><tr><th>Name</th><th>Description</th></tr><tr><td>DBC_CCV_W_TrBrEff</td><td>Traction brake effort request</td></tr><tr><td>DBC_BSV_XF_IduAchTrEffCst</td><td>Sum of achieved traction effort</td></tr><tr><td>DBC_BSV_XF_IduAchEdBrCst</td><td>Sum of achieved ED brake effort</td></tr><tr><td>DBC_BSV_XF_BcuBrAchCst</td><td>Sum of achieved DEP brake effort</td></tr><tr><td>DBC_CCT_X_SpdMdRef</td><td>Train speed reference</td></tr><tr><td>DBC_CCV_Xv_SpdLim</td><td>Speed limitation</td></tr><tr><td>MPW_BCV_S_LnTrp_T4</td><td>Status, line trip</td></tr><tr><td>MPW_BCV_S_LnTrpCmpl_T4</td><td>Status, line trip complete</td></tr><tr><td>MPW_BCV_C_TstMdLnTrpTst_T4</td><td>Test mode, line trip test</td></tr><tr><td>MPW_BCV_S_LnTrpOk_T4</td><td>Status, line trip test is OK</td></tr><tr><td>MPW_BCV_S_LimTstToUnt1</td><td>Status: Line trip test Time out</td></tr><tr><td>MPW_BCV_S_LimTstWaUnt1</td><td>Status: Line trip test Warning</td></tr><tr><td>TC_BI_1PCU1_VU_LcmAcLnRmSys</td><td>Valid bit: LCM AC line voltage in T4 from PCU1</td></tr><tr><td>TC_BI_3PCU2_VU_LcmAcLnRmSys</td><td>Valid bit: LCM AC line voltage in T4 from PCU3</td></tr><tr><td>TC_BI_1PCU1_S_HVMst</td><td>Status: Local PCU1 is High Voltage Master, Unit1</td></tr><tr><td>TC_BI_3PCU2_S_HVMst</td><td>Status: Local PCU3 is High Voltage Master, Unit1</td></tr><tr><td>TC_BI_1PCU1_XU_LcmAcLnRmSys</td><td>LCM AC line voltage in T4 from PCU1</td></tr><tr><td>TC_BI_3PCU2_XU_LcmAcLnRmSys</td><td>LCM AC line voltage in T4 from PCU3in</td></tr><tr><td>MPW_BCV_S_Pcu1LcbOpd</td><td>LCB is opened from PCU1</td></tr><tr><td>MPW_BCV_S_Pcu1LcbCld</td><td>LCB is closed from PCU1</td></tr><tr><td>MPW_BCV_S_Pcu3LcbOpd</td><td>LCB is opened from PCU3</td></tr><tr><td>MPW_BCV_S_Pcu3LcbCld</td><td>LCB is closed from PCU3</td></tr></table> | PCU ><br>CCU-0 | Signal                                         | Cycle<br>time | Description | Scale | CX1CT807 | HCV_XU_AcLn | 64 ms | LCM AC line voltage to remote system | 1=10V | CX1CT8027 | HCV_XU_AcLn | 64 ms | Valid bit LCM AC line voltage to remote system | L | Name | Description | DBC_CCV_W_TrBrEff | Traction brake effort request | DBC_BSV_XF_IduAchTrEffCst | Sum of achieved traction effort | DBC_BSV_XF_IduAchEdBrCst | Sum of achieved ED brake effort | DBC_BSV_XF_BcuBrAchCst | Sum of achieved DEP brake effort | DBC_CCT_X_SpdMdRef | Train speed reference | DBC_CCV_Xv_SpdLim | Speed limitation | MPW_BCV_S_LnTrp_T4 | Status, line trip | MPW_BCV_S_LnTrpCmpl_T4 | Status, line trip complete | MPW_BCV_C_TstMdLnTrpTst_T4 | Test mode, line trip test | MPW_BCV_S_LnTrpOk_T4 | Status, line trip test is OK | MPW_BCV_S_LimTstToUnt1 | Status: Line trip test Time out | MPW_BCV_S_LimTstWaUnt1 | Status: Line trip test Warning | TC_BI_1PCU1_VU_LcmAcLnRmSys | Valid bit: LCM AC line voltage in T4 from PCU1 | TC_BI_3PCU2_VU_LcmAcLnRmSys | Valid bit: LCM AC line voltage in T4 from PCU3 | TC_BI_1PCU1_S_HVMst | Status: Local PCU1 is High Voltage Master, Unit1 | TC_BI_3PCU2_S_HVMst | Status: Local PCU3 is High Voltage Master, Unit1 | TC_BI_1PCU1_XU_LcmAcLnRmSys | LCM AC line voltage in T4 from PCU1 | TC_BI_3PCU2_XU_LcmAcLnRmSys | LCM AC line voltage in T4 from PCU3in | MPW_BCV_S_Pcu1LcbOpd | LCB is opened from PCU1 | MPW_BCV_S_Pcu1LcbCld | LCB is closed from PCU1 | MPW_BCV_S_Pcu3LcbOpd | LCB is opened from PCU3 | MPW_BCV_S_Pcu3LcbCld | LCB is closed from PCU3 | Derived<br>Requirement | 4S_09.03.05.-.TCMS<br>Software |  | SIL 0 |
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| PCU ><br>CCU-0                          | Signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| CX1CT807                                | HCV_XU_AcLn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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        |       |           |             |       |                                                |   |      |             |                   |                               |                           |                                 |                          |                                 |                        |                                  |                    |                       |                   |                  |                    |                   |                        |                            |                            |                           |                      |                              |                        |                                 |                        |                                |                             |                                                |                             |                                                |                     |                                                  |   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| CX1CT8027                               | HCV_XU_AcLn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 64 ms          | Valid bit LCM AC line voltage to remote system | L             |             |       |          |             |       |                                      |       |           |             |       |                                                |   |      |             |                   |                               |                           |                                 |                          |                                 |                        |                                  |                    |                       |                   |                  |                    |                   |                        |                            |                            |                           |                      |                              |                        |                                 |                        |                                |                             |                                                |                             |                                                |                     |                                                  |   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| Name                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| DBC_CCV_W_TrBrEff                       | Traction brake effort request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| DBC_BSV_XF_IduAchTrEffCst               | Sum of achieved traction effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| DBC_BSV_XF_IduAchEdBrCst                | Sum of achieved ED brake effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| DBC_BSV_XF_BcuBrAchCst                  | Sum of achieved DEP brake effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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| DBC_CCT_X_SpdMdRef                      | Train speed reference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| DBC_CCV_Xv_SpdLim                       | Speed limitation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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| MPW_BCV_S_LnTrp_T4                      | Status, line trip                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| MPW_BCV_S_LnTrpCmpl_T4                  | Status, line trip complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| MPW_BCV_C_TstMdLnTrpTst_T4              | Test mode, line trip test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| MPW_BCV_S_LnTrpOk_T4                    | Status, line trip test is OK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| MPW_BCV_S_LimTstToUnt1                  | Status: Line trip test Time out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| MPW_BCV_S_LimTstWaUnt1                  | Status: Line trip test Warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| TC_BI_1PCU1_VU_LcmAcLnRmSys             | Valid bit: LCM AC line voltage in T4 from PCU1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| TC_BI_3PCU2_VU_LcmAcLnRmSys             | Valid bit: LCM AC line voltage in T4 from PCU3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| TC_BI_1PCU1_S_HVMst                     | Status: Local PCU1 is High Voltage Master, Unit1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| TC_BI_3PCU2_S_HVMst                     | Status: Local PCU3 is High Voltage Master, Unit1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| TC_BI_1PCU1_XU_LcmAcLnRmSys             | LCM AC line voltage in T4 from PCU1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| TC_BI_3PCU2_XU_LcmAcLnRmSys             | LCM AC line voltage in T4 from PCU3in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| MPW_BCV_S_Pcu1LcbOpd                    | LCB is opened from PCU1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| MPW_BCV_S_Pcu1LcbCld                    | LCB is closed from PCU1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| MPW_BCV_S_Pcu3LcbOpd                    | LCB is opened from PCU3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |
| MPW_BCV_S_Pcu3LcbCld                    | LCB is closed from PCU3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                  |                                                  |                             |                                     |                             |                                       |                      |                         |                      |                         |                      |                         |                      |                         |                        |                                |  |       |

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### 3.4.3 Isolate High Voltage

#### 3.4.3.1 Function Design

##### 3.4.3.1.1 Normal Condition

It is possible to isolate short circuits or failures to keep half of the traction of an 8 car train available.

##### 3.4.3.1.2 Failure Condition

- 1) One disconnector does not open on order.
- 2) One disconnector does not close on order.
- 3) Position of disconnectors not known.

##### 3.4.3.1.3 Failure Supervision

- 1) Roof line disconnector does not open on order. Event will be set and driver must change pantograph manually.
- 2) Roof line disconnector does not close on order. Event will be set and driver must change pantograph manually.
- 3) Roof line disconnector is in an unknown position. Event will be set and driver must change pantograph manually.

#### 3.4.3.2 Requirements

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.474                                  |
| <b>Description</b>  | It shall be possible to isolate separate high voltage sections. |
| <b>Safety level</b> | SIL 0                                                           |

A train half is considered "isolated from DC catenary" when the DC line disconnector related to active pantograph is open and either its DC pantographs are lowered or the system isolating switch is open.

A train half is considered "isolated from AC catenary" when the AC line disconnector related to active pantograph is open and either its AC pantographs are lowered or the AC LCB is open.

| Nr                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Applicability Details | Derived to           | User Interface | SIL |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.997 | <p>The roof line disconnector shall be considered to be not opening on order if both feedbacks indicate closed position after:</p> <ul style="list-style-type: none"> <li>– 3 seconds for AC disconnector</li> <li>– 1 second for DC disconnector</li> </ul> <p>Three attempts shall be allowed to open with 10 seconds interval between each attempt. If after 3 attempts the roof line disconnector has not opened on order the roof line disconnector shall be indicated as faulty to TCMS.</p> | Derived Requirement   | 4S_04.-.Power Supply |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.998 | <p>The roof line disconnector shall be considered to be not closing on order if both feedbacks indicating open position after:</p> <ul style="list-style-type: none"> <li>– 3 seconds for AC disconnector</li> <li>– 1 second for DC disconnector</li> </ul> <p>Three attempts shall be allowed to close with 10 seconds interval between each attempt. If after 3 attempts the disconnector has not closed on order then both disconnectors shall be ordered to open.</p>                         | Derived Requirement   | 4S_04.-.Power Supply |                | N/A |

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| 2F_04.01.Zefiro-Europe.CONCEPT.999  | The roof line disconnecter shall be considered to be in an unknown position if neither open or close acknowledge is set or open and close acknowledge are set simultaneously for 4 seconds. Three attempts shall be allowed to open with 10 seconds interval between each attempt. If the position is still unknown after 3 attempts roof line disconnecters shall be indicated as faulty to TCMS. | Derived Requirement | 4S_04.-.Power Supply        |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1000 | If the roof line disconnecter (only in AC mode) is indicated as faulty from the PCU (propulsion control unit) then the pantograph related to the faulty roof line disconnecter shall be blocked and an event shall be set to indicate to the driver that pantograph change is required.<br><br>Note: the behaviour here above is valid only for AC mode                                            | Derived Requirement | 4S_09.03.05.-.TCMS Software |     | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1222 | If the system isolation switch is indicated as faulty from the PCU (propulsion control unit) then the pantograph related to the faulty switch shall be blocked and an event shall be set to indicate to the driver that pantograph change is required.                                                                                                                                             | Derived Requirement | 4S_09.03.05.-.TCMS Software |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.474  | The roof line disconnecters shall be closed at startup of train                                                                                                                                                                                                                                                                                                                                    | Derived Requirement | 4S_04.-.Power Supply        |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1223 | The TCMS, since the start up of the train till the shut down, has to send the status of the battery contactor closed to the propulsion system.                                                                                                                                                                                                                                                     | Derived Requirement | 4S_09.03.05.-.TCMS Software |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.475  | The status (open/closed) of the roof line disconnecters shall be displayed on the IDU.                                                                                                                                                                                                                                                                                                             | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.137  | If a failure in HV-part is detected, which requires separation of both halves of the train (e.g. earth fault), the roof line disconnecter related to active pantograph shall be requested to open.                                                                                                                                                                                                 | Derived Requirement | 4S_04.-.Power Supply        |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.467  | The DC-line disconnecters shall only be operated when all pantographs are lowered or DC LCBs are opened.                                                                                                                                                                                                                                                                                           | Derived Requirement | 4S_04.-.Power Supply        |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.586  | The AC-line disconnecter shall only be operated when all AC pantographs are lowered or all AC LCBs are in open position.                                                                                                                                                                                                                                                                           | Derived Requirement | 4S_04.-.Power Supply        |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1226 | the IDU has to provide a cut in/out virtual button for the driver to request the manual opening of the the roof disconnectors 3 or 25 kV. This button must be sleevtive for train and line voltage.                                                                                                                                                                                                | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1227 | When the TCMS recives the roof disconnecter cut out request, the cut out request has to be sent to propulsion only if:<br>– Train is standstill<br>AND<br>– All panto are down OR All LCB are open                                                                                                                                                                                                 | Derived Requirement | 4S_09.03.05.-.TCMS Software |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1228 | When the roof disconnecter cut out request is ordered only the selected voltage (TCMS has to check the driver desk selector) roof disconnecter shall be opened.<br><br>Note: if 3kv is selected and the driver request the roof disconnecter opening only the DC disconnecter request has to be sent to propulsion (see icd variable for request)                                                  | Derived Requirement | 4S_09.03.05.-.TCMS Software |     | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1231 | When the roof disconnecter opening request is ordered from the driver the TCMS implements the following actions:<br>– a Master reset is sent to all equipments<br>– 10 s after the master reset the TCMS request to propulsion to open both (3 kV or 25kV) roof disconnecter                                                                                                                       | Derived Requirement | 4S_09.03.05.-.TCMS Software |     | N/A   |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1229 | In case the roof disconnecter opening order is triggered to propulsion an event shall be set for engineering                                                                              | Derived Requirement | 4S_09.03.05.-TCMS Software | EVENT | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1230 | In case the the driver request a roof disconnecter opening but not all conditions are true a selective (one event for each condition not verified) event shall be set to the driver       | Derived Requirement | 4S_09.03.05.-TCMS Software | EVENT | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1232 | The roof disconnectors can be cut in if:<br>– a cut out command has been previously delivered<br>AND<br>– Train is standstill<br>AND<br>– All panto are down<br>AND<br>– All LCB are open | Derived Requirement | 4S_09.03.05.-TCMS Software |       | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1233 | In case the roof disconnecter closing order is triggered to propulsion an event shall be set for engineering                                                                              | Derived Requirement | 4S_09.03.05.-TCMS Software | EVENT | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1234 | In case the the driver request a roof disconnecter closing but not all conditions are true a selective (one event for each condition not verified) event shall be set to the driver       | Derived Requirement | 4S_09.03.05.-TCMS Software | EVENT | N/A |

### 3.5 Energy Meter

The energy metering function is a single sub-system function and works independent from any vehicle level function. The vehicle only provides auxiliary power from the DC110V level and generates events if the power supply is lost. Additionally vehicle can provide the location and time data to the system and send the data generated by Energy Meter to the ground station.

#### 3.5.1 Function Design

##### Energy Meter System:

- is a combination of the Energy Metering Function (EMF), Data Handling System (DHS), Communication Unit (GSM-R / 3G /4G) and Global Positioning System (GPS).
- It shall be approved for billing by Eress.
- It shall be capable of measuring the energy absorbed (consumed) and regenerated by trains.
- The energy data is time-integrated based on a reference period which is selectable and stored on the device internal memory for at least 60 days.
- The vehicle GPS location, date and time data can be correlated with stored energy data.
- Energy Data can be shared with TCMS including real time voltage, current and power readings.
- The diagnostic data is shared with TCMS.
- Vehicle forwards the stored energy data from the energy meter system (DHS) to the ground station (Data Collection System DCS).

#### 3.5.2 Requirements

|                     |                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.796                                                                                  |
| <b>Description</b>  | The vehicle shall provide the location data and time data to the Energy meter system (DHS) for billing purpose. |
| <b>Safety level</b> | SIL 0                                                                                                           |

| Nr                                  | Description                                                                                                                                                                        | Applicability Details | Derived to                 | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1313 | <u>TCMS: send location and time data to EMS</u><br>GIVEN the communicaton is established between TCMS and EMS<br>THEN TCMS shall send the location data and time date data to EMS. | Derived Requirement   | 4S_09.03.05.-TCMS Software |                | SIL 0 |

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|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.797                                                                                                  |
| <b>Description</b>  | The vehicle shall forward the energy data from the Energy meter system (DHS) to the ground station Data Collection System DCS). |
| <b>Safety level</b> | SIL 0                                                                                                                           |

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                               | Applicability Details | Derived to           | User Interface | SIL   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1315 | <u>EMS: send energy data to the ground station (DCS)</u><br>GIVEN <ul style="list-style-type: none"> <li>communication to the ground station (Data Collection System) is established AND</li> <li>energy data needs to be sent to the ground station (DCS)</li> </ul> THEN EMS shall send the energy data to the TCMS to forward to ground station (DCS). | Derived Requirement   | 4S_04.-.Power Supply |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1319 | <u>EMS: send diagnostic data to TCMS</u><br>GIVEN the EMS is supplied with DC110V supply<br>THEN EMS shall send the diagnostic data to TCMS.                                                                                                                                                                                                              | Derived Requirement   | 4S_04.-.Power Supply |                | SIL 0 |

|                     |                                                                      |
|---------------------|----------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.798                                       |
| <b>Description</b>  | The vehicle shall manage data from/to the Energy meter system (DHS). |
| <b>Safety level</b> | SIL 0                                                                |

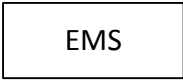
| Nr                                  | Description                                                                                                                                                                                                                        | Applicability Details | Derived to                                                  | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1343 | GIVEN MCG is active<br>THEN MCG shall foreseen a transparent channel for wayside communication of EMS in accordance with EN 50463-1:2017                                                                                           | Derived Requirement   | 4S_09.03.01.01.-<br>.Train to Wayside Communication Systems |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1345 | GIVEN TCSM is active<br>THEN TCMS shall transmit data with EMS according ICD [1]                                                                                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software                                 |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1347 | GIVEN LCB in T4 is closed AND<br>LCB in T5 is open<br>THEN EMS shall be able to receive data from sensors installed in T4 AC HIGH VOLTAGE BOX instead of data from T5 AC HIGH VOLTAGE BOX                                          | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits     |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1348 | GIVEN LCB in T4 is closed AND<br>LCB in T5 is open<br>THEN TCMS shall energize JC-K001 relay so that EMS will be able to receive data from sensors installed in T4 AC HIGH VOLTAGE BOX instead of data from T5 AC HIGH VOLTAGE BOX | Derived Requirement   | 4S_09.03.05.-.TCMS Software                                 |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1349 | GIVEN LCB in T4 is closed AND<br>LCB in T5 is open<br>WHEN JC-K001 relay is not energized<br>THEN TCMS shall set a diagnostic event which advises that EMS is not able to receive data from T4 AC HIGH VOLTAGE BOX                 | Derived Requirement   | 4S_09.03.05.-.TCMS Software                                 | EVENT          | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1350 | GIVEN LCB in T4 is closed AND<br>LCB in T5 is open<br>WHEN JC-K001 relay is energized<br>THEN TCMS shall reset the diagnostic event which advises that EMS is not able to receive data from T4 AC HIGH VOLTAGE BOX                 | Derived Requirement   | 4S_09.03.05.-.TCMS Software                                 | EVENT          | SIL 0 |

|                     |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.799                                                             |
| <b>Description</b>  | Status of Energy meter system (DHS) and relevant diagnostic events shall be showed on IDU. |
| <b>Safety level</b> | SIL 0                                                                                      |

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| Nr                                  | Description                                                                                                                                                                                                   | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1352 | GIVEN TCMS is active<br>WHEN EMS is not in Failure<br>THEN TCMS shall show on IDU the status of EMS<br><br><EMS icon><br>    | Derived Requirement   | 4S_09.03.05.-.TCMS Software | HW HMI         | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1353 | GIVEN TCMS is active<br>WHEN EMS is in Failure<br>THEN TCMS shall show on IDU the status of EMS<br><br><Faulty EMS icon><br> | Derived Requirement   | 4S_09.03.05.-.TCMS Software | HW HMI         | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1355 | GIVEN TCMS is active<br>WHEN EMS is in Failure<br>THEN TCMS shall trigger relevant diagnostic events                                                                                                          | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | SIL 0 |

### 3.6 Key interlocking

The Key Interlocking System is described in the document 3EST000226-2279.

#### 3.6.1 Function Design

##### 3.6.1.1 Normal Condition

All power supply available: HV, 400V, 110V

##### 3.6.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy\_XXX.xls”.

##### 3.6.1.3 Failure Supervision

Status of HV-switches and disconnectors is controlled by TCMS and hardwired to control lamp and solenoid protected lock.

#### 3.6.2 Requirements

|                     |                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.679                                                               |
| <b>Description</b>  | Two levels of key interlocking shall be supported:<br>– Operation mode<br>– Maintenance mode |
| <b>Safety level</b> | SIL 2                                                                                        |

At Interlocking sequence there are five different modes defined:

- Operation Mode
- Parking Mode
- Disabled Mode
- Depot Supply Mode
- Maintenance Mode

|                             |                  |                  |              |                        |
|-----------------------------|------------------|------------------|--------------|------------------------|
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|                     |                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.680                                                                                                                |
| <b>Description</b>  | In operation mode, high voltage components and auxiliary power supply shall be locked to prevent unwanted and dangerous switching operations. |
| <b>Safety level</b> | SIL 2                                                                                                                                         |

|                     |                                                                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.681                                                                                                                                                                                                                                       |
| <b>Description</b>  | Changing from "operation" to "maintenance" mode, the following steps shall be performed: <ul style="list-style-type: none"> <li>– open line circuit breakers</li> <li>– lowering pantographs</li> <li>– earthing the high voltage equipment, if necessary</li> </ul> |
| <b>Safety level</b> | N/A                                                                                                                                                                                                                                                                  |

- 1) Operation Mode: HV is live, active MCB is closed, active pantograph is raised, internal 3AC400V is live, 110 VDC is live, depot supply is not possible. KIS is not active.
- 2a) Parking Mode: is identical with Operation Mode, but the **traction is blocked and the Parking brake is applied**.
- 2b) Disabled Mode: **HV shut off, all MCBs are open, all pantographs are down, both Isolating/Grounding Switches DC (4a) are in earthed position, depot supply is possible, contactors for 3AC400V are open**, and 110 VDC remains live. Disabled Mode is a parking mode for the train with the possibility to connect depot supply; e.g. for cleaning the train with a vacuum cleaner or charging the 110V batteries. To achieve this mode, the KIS is not active.
- 3) Depot Supply Mode: HV shut off **and earthed**, all pantographs are down, depot supply is possible, 110 VDC remains live for a limited time; **access to the HV boxes at the roof is possible**. Depot Supply Mode is a mode, where maintenance work can be performed at cubicles containing live 3AC400V equipment. Also maintenance at roof equipment is possible. To achieve this mode, the KIS is needed.
- 4) Maintenance Mode: HV shut off and earthed, pantographs are down, **3AC400 V is shut off and earthed, DC-links are earthed; access to the key-locked cubicles is possible**, 110 VDC remains live, **depot supply is not possible**. Maintenance Mode is a mode, where maintenance work can be performed at cubicles containing HV equipment or for exchanging 3AC400V equipment. To achieve this mode, the KIS is needed.

| Nr                                 | Description                                                                 | Applicability Details | Derived to                  | User Interface | SIL   |
|------------------------------------|-----------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.602 | If any A-Key is activated the LCB shall open and the pantograph shall lower | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

|                     |                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.682                                                            |
| <b>Description</b>  | The hatches of the converter unit must be locked to prevent access to dangerous voltages. |
| <b>Safety level</b> | SIL 2                                                                                     |

### 3.6.3 Function Earthing of High Voltage Supply

#### 3.6.3.1 Function Design

##### 3.6.3.1.1 Normal Condition

The High Voltage system is earthed to secure that maintenance of the system can be done in a secure way without risk for electric shock.

##### 3.6.3.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File "FMECA KIS Zefiro Italy\_XXX.xls".

- 1) Isolating/Grounding Switch DC (4a) does not close
- 2) Isolating/Grounding Switch AC (4b) does not close

|                                    |                         |                         |                     |                        |
|------------------------------------|-------------------------|-------------------------|---------------------|------------------------|
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### 3.6.3.1.3 Failure Supervision

- 1) HV DC line is earthed manually by HV DC Earthing Disconnectors 36a and 36b.
- 2) Status of Isolating/Grounding Switch AC is controlled by PCU (via TCMS) and hardwired to a control lamp and a solenoid protected lock

### 3.6.3.2 Requirements

|                     |                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.291                                                                                                                                                                                                              |
| <b>Description</b>  | It shall be possible to earth the high voltage supply with a switch remotely controlled. Earthing of the high voltage AC supply shall be done after the line circuit breaker. With this measure the main transformer shall also be earthed. |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                       |

| Nr                                  | Description                                                                                                                                                                                                                                    | Applicability Details | Derived to                                              | User Interface | SIL   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1016 | When the push-button "Lamp Test" from the A1-key Panel in DM1-car is pressed the Blue lamp "HV AC earthed" at A1-Key-Panel will light.                                                                                                         | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1017 | When the push-button "Lamp Test" from the A2-key Panel in DM8-car is pressed the Blue lamp "HV AC earthed" at A2-Key-Panel will light.                                                                                                         | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.147  | When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position the AC- and DC-LCBs shall be ordered to open and all pantographs shall be lowered.                                                                      | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1060 | When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position the AC- and DC-LCBs shall be ordered to open and all pantographs shall be lowered.                                                                      | Derived Requirement   | 4S_04.-.Power Supply                                    |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1018 | When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position all 3AC400V contactors 18a, 18c and 39 shall open.                                                                                                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1078 | When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position all 3AC400V contactors 18a shall open.                                                                                                                  | Derived Requirement   | 4S_05.-.Propulsion                                      |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1218 | when the A-key is restored to 'normal mode' position (from 'Depot mode') all MV contactors 18c (medium voltage contactor 7) and 39 (medium voltage contactor 6) and also the medium voltage contactor 8 shall be closed.                       | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1040 | When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position pre charging devices 29a, 29b, 32a, 32b and disconnectors 33a, 33b shall open.                                                                          | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1079 | When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position pre charging devices 29a, 29b, 32a, 32b and disconnectors 33a, 33b shall open.                                                                          | Derived Requirement   | 4S_05.-.Propulsion                                      |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1019 | When the A1-Key at A1-Key Panel in DM1-car or the A2-Key at A2-Key Panel in DM8-car is moved to the Depot-Supply position the Grounding switches 4a ( TT2 and TT7-car) and 4b (T4 and T5-car) shall be ordered to close the earthing contacts. | Derived Requirement   | 4S_09.03.05.-.TCMS Software                             |                | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1080 | When the A1-Key at A1-Key Panel in DM1-car or the A2-Key at A2-Key Panel in DM8-car is moved to the Depot-Supply position the Grounding switches 4a ( TT2 and TT7-car) and 4b (T4 and T5-car) shall be ordered to close the earthing contacts. | Derived Requirement   | 4S_04.-.Power Supply                                    |                | SIL 2 |

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|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------|-------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1020 | When the A1-Key or the A2-Key is moved to the Depot-Supply position the Blue lamps "HV AC earthed" at both A-Key-Panels in DM1- and DM8-car will light, if Grounding switches AC 4b in T4- and T5-car are in the earthed position.                                                                                                                                                                                                                                                                                                                          | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1021 | When the A1-Key or the A2-Key is moved to the Depot-Supply position the solenoid protected locks for the A-Keys at HV DC Earthing Disconnector 36a and 36b (TT2- and TT7-car) will be unlocked, if Grounding switches AC 4b in T4- and T5-car are in the earthed position.                                                                                                                                                                                                                                                                                  | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.413  | The status of the Grounding switches 4a and 4b shall be displayed on the IDU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software                             | IDU   | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.414  | All orders for earthing shall be given simultaneously to all HV Grounding switches 4a and 4b on the train.                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Derived Requirement | 4S_09.03.05.-.TCMS Software                             |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.417  | The close order signal shall be set low as soon as the feedback Grounding switch 4a and 4b CLOSED goes high.                                                                                                                                                                                                                                                                                                                                                                                                                                                | Derived Requirement | 4S_04.-.Power Supply                                    |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.419  | The open order signal shall be set low as soon as the feedback Grounding switch 4a and 4b OPEN goes high.                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Derived Requirement | 4S_04.-.Power Supply                                    |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.420  | The maximum time for close or open order shall be 4 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Derived Requirement | 4S_04.-.Power Supply                                    |       | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.421  | If the A-key has not been turned to the Depot - Supply position for >7days, then there shall be an event displayed on the IDU. The event shall be reset when the key has been in the Depot - Supply position and the Grounding switches 4a (open when earthed) and 4b (closed when earthed) have followed the right sequence when the A-key is returned to the normal position (4a remains open till pantograph command, 4b is open when not earthed). This ensures that the key is operated to test the Grounding switches function at least every 7 days. | Derived Requirement | 4S_09.03.05.-.TCMS Software                             | EVENT | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.422  | The Grounding Switches 4b shall not be able to close when any pantograph is raised.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Derived Requirement | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |       | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.755  | The Grounding Switch 4b shall not be ordered to close when any pantograph is raised.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement | 4S_04.-.Power Supply                                    |       | SIL 0 |
| 2F_04.01.Zefiro-Europe.CONCEPT.412  | When the input from the A-key panel in both DM-cars is not in the Depot- Supply position the Grounding switches 4b (T4 and T5-car) shall be ordered to open.                                                                                                                                                                                                                                                                                                                                                                                                | Derived Requirement | 4S_09.03.05.-.TCMS Software                             |       | SIL 0 |

#### Earth HV DC line manually (left side of the train):

Actions: Manual actions

- Remove A1-key from A1-key panel and put it into the lock of HV DC Earthing Disconnector panel switch 36a (Z-switches in car TT2) and turn it.
- Turn 36a (Z-switches) into earthing position

Results: mechanical interlocking

- The left DC-HV box is earthed by disconnector 36a (Z-switches) manually
- One E1-keys is released for turning and taking off; mechanically
- One B1-key is released for turning and taking off; mechanically

The A1-key cannot be removed if the switch 36a is in earthed position; mechanically

| Nr                                  | Description                                                                                                          | Applicability Details | Derived to                                              | User Interface | SIL   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1024 | The position of HV DC Earthing Disconnector 36a (Z-switches) shall be monitored by TCMS with two auxiliary contacts. | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 0 |

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### Earth HV DC line manually (right side of the train):

Actions: Manual actions

- Remove A2-key from A2-key panel and put it into the lock of HV DC Earthing Disconnecter panel switch 36b (Z-switches in car TT7) and turn it.
- Turn 36b (Z-switches) into earthing position

Results: mechanical interlocking

- The right DC-HV box is earthed by disconnecter 36b (Z-switches) manually
- One E2-keys is released for turning and taking off; mechanically
- One B2-key is released for turning and taking off; mechanically

The A2-key cannot be removed if the switch 36b is in earthed position; mechanically

| Nr                                  | Description                                                                                                          | Applicability Details | Derived to                                              | User Interface | SIL   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1026 | The position of HV DC Earthing Disconnecter 36b (Z-switches) shall be monitored by TCMS with two auxiliary contacts. | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 0 |

## 3.6.4 Function Earthing of Converter boxes

### 3.6.4.1 Function Design

#### 3.6.4.1.1 Normal Condition

The High Voltage system is earthed to secure that maintenance of the system can be done in a secure way without risk for electric shock.

#### 3.6.4.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy\_XXX.xls”.

- 1) The earthing switch does not close on order, that is the system is not earthed

#### 3.6.4.1.3 Failure Supervision

- 1) Key to converter unit can not be released if the earthing switch is not in earthed position.

### 3.6.4.2 Requirements

|                     |                                                                                                                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>           | 2F_04.01.Zefiro-Europe.TRS.683                                                                                                                                                                                                                                                                                |
| <b>Description</b>  | An earth switch shall be installed near each converter unit. The earth switch is operated to earth the internal connections (DC-link) of the converter unit in maintenance mode. In order to access the converter unit hatches, the earth switch shall be operated, which gives access to a key to the hatch. |
| <b>Safety level</b> | SIL 2                                                                                                                                                                                                                                                                                                         |

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                 | Applicability Details | Derived to                  | User Interface | SIL   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.151  | An earth switch shall be installed on the DC link of the converter to discharge all capacitors before the maintenance on the converter units.                                                                                                                                                                                                                                                                               | Derived Requirement   | 4S_05.-.Propulsion          |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1054 | <b>Blocking 3AC400V (right side and left side of the train):</b><br>Actions: manual action <ul style="list-style-type: none"> <li>• Turn B1-key at the HV DC Earthing Disconnecter panel 36a</li> </ul> Results: <ul style="list-style-type: none"> <li>• Blocking 3AC400V will be initiated by opening 18b (Depot Supply Contactor) <b>on both sides of the train.</b></li> </ul> The B1-key can be released. mechanically | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | SIL 0 |

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|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|--|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1055 | <b>Blocking 3AC400V (right side and left side of the train):</b><br>Actions: manual action <ul style="list-style-type: none"> <li>Turn B2-key at the HV DC Earthing Disconnecter panel 36b</li> </ul> Results: <ul style="list-style-type: none"> <li>Blocking 3AC400V will be initiated by opening 18b (Depot Supply Contactor) <b>on both sides of the train.</b></li> </ul> The B2-key can be released. mechanically | Derived Requirement | 4S_09.03.05.-.TCMS Software |  | SIL 0 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|--|-------|

#### Earthing 3AC400V (left side of the train):

Actions: manual action

- take B1-key, put it into the 400V Earthing Disconnecter 37a of car T4, and turn it.
- Turn handle of 37a and earth.

Results:

- The B1-key cannot be removed; mechanically
- two C1-keys will be released; mechanically

| Nr                                  | Description                                                                                            | Applicability Details | Derived to                                          | User Interface | SIL |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.1030 | The position of 400V Earthing Disconnecter 37a shall be monitored by TCMS with two auxiliary contacts. | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1081 | When the 400V Earthing Disconnecter is closed it shall be displayed on IDU.                            | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         | IDU            | N/A |

#### Earthing 3AC400V (right side of the train):

Actions: manual action

- take B2-key, put it into the 400V Earthing Disconnecter 37b of car T5, and turn it.
- Turn handle of 37b and earth.

Results:

- The B2-key cannot be removed; mechanically
- two C2-keys will be released; mechanically

| Nr                                  | Description                                                                                            | Applicability Details | Derived to                                          | User Interface | SIL   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1031 | The position of 400V Earthing Disconnecter 37b shall be monitored by TCMS with two auxiliary contacts. | Derived Requirement   | 4S_09.02.09.-.Conventional Train Control - Circuits |                | N/A   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1057 | When the 400V Earthing Disconnecter is closed it shall be displayed on IDU.                            | Derived Requirement   | 4S_09.03.05.-.TCMS Software                         | IDU            | SIL 0 |

#### Earthing the DC-links (left side, cars DM1 and M3):

Actions: manual action

- Take the two C1-keys, put them into the locks of the two DC-Link Earthing Disconnectors 35a (cars DM1 and M3), and turn them.
- Turn handles of the earthing disconnectors and earth.

Results:

- DC links and HV path (step down chopper (11)) are earthed.
- The C1-keys cannot be released; mechanically
- Two D1-keys from car DM1 and two D2-keys from car M3 are released; mechanically

#### Earthing the DC-links (right side, cars DM8 and M6):

Actions: manual action

- Take the two C2-keys, put them into the locks of the two DC-Link Earthing Disconnectors 35b (cars DM8 and M6), and turn them.

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- Turn handles of the earthing disconnectors and earth.

Results:

- DC links and HV path (step down chopper (11)) are earthed.
- The C2-keys cannot be released; mechanically
- Two D3-keys from car DM8 and two D4-keys from car M6 are released; mechanically

#### Getting the F-Keys (left side)

Actions: left side, at car DM1; manual action

- Take one of the D1-, D2-, D3- and D4-keys and put all four keys into the F Key Multiplier at car DM1
- Turn them all

Result:

- 12 F-keys are released
- The D1-, D2-, D3- and D4-keys cannot be released, if at least one F-key is taken off.

#### Getting the F-Keys (right side)

Actions: right side, at car DM8; manual action

- Take one of the D1-, D2-, D3- and D4-keys and put all four keys into the F Key Multiplier at car DM8
- Turn them all

Result:

- 12 F-keys are released
- The D1-, D2-, D3- and D4-keys cannot be released, if at least one F-key is taken off.

#### Bringing back the train into Operation Mode

The sequence of the KIS has to be performed into the reverse direction for each side of the train.

First all hatches protected by the F-keys have to be closed. The F-keys have to be taken back into its key multipliers in order to get access to the D1-, D2-, D3 and D4-keys. (mechanically)

The D1-, D2-, D3- and D4-keys have to be brought back into the DC-Link Earthing Disconnectors 35a+b, the earthing handles have to be brought into the "not earthed" position and then the C1- and C2-keys are released. (manually)

With the two C1-keys (and the two C2-keys) the 400V Earthing Disconnectors 37a+b for 3AC400V are reset by switching into the "not earthed" position. So, the B1- and B2-keys are released. (manually)

The B1/B2-keys and the E1/E2-keys have to be brought back into the disconnector panels 36a+b and turned, and then switches 36a+b must be turned into the "not earthed" position which will release the A1 and A2-keys. (manually)

Now we are back in Operation Mode.

The A1/A2-keys must be brought back into the A-key panels 40a/b and turned. This opens the Grounding switches 4b from AC-MCBs in cars T4 and T5.

| Nr                                  | Description                                                                                                                                                                                        | Applicability Details | Derived to                                              | User Interface | SIL   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|----------------|-------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1038 | After opening of the Grounding switches 4b (left and right AC-MCB) the solenoids in the HV DC Earthing Disconnectors 36a and 36b must switched off and the lock from A1-Key and A2-Key is blocked. | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | SIL 2 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1039 | After opening of the Grounding switches 4b (left and right AC-MCB ) the Blue lamps "AC HV earthed" at A1- and A2-Key-Panel will switch off.                                                        | Derived Requirement   | 4S_09.02.09.-<br>.Conventional Train Control - Circuits |                | N/A   |

Now the Disabled Mode is achieved.

To bring the train into Parking Mode the Tumbler Switches "pantograph up" and "MCB closed" have to operated.

From Parking Mode the train can be brought back into Operation Mode by releasing the parking brakes and releasing the traction control.

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### 3.7 HVAC pre-conditioning on DC line

A Pre-Conditioning (PC) functionality shall be foreseen on DC lines in order to allow train to supply all the needed power to HVAC system without exceeding current limitation (200 Amps per pantograph on 3kVdc, 300 Amps per pantograph on 1.5Vdc networks)

Pre conditioning functionality will be requested to HVAC under DC line voltage only when requested by the driver. On AC lines instead, PC mode will be automatically managed by TCMS

#### 3.7.1 Requirements

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Applicability Details | Derived to                  | User Interface | SIL |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.1192 | A dedicated menu on IDU shall be foreseen to enable Pre-Conditioning function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Derived Requirement   | 4S_09.03.05.-.TCMS Software | IDU            | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1193 | TCMS shall check that all following condition are true before let the driver enable PC mode: <ul style="list-style-type: none"> <li>– Train standstill</li> <li>– Parking brake applied</li> <li>– Any DC system is selected (3kV or 1.5kV)</li> <li>– Both pantographs are down (and all LCB open)</li> <li>– NO panto or LCB are set in faulty/cut-out status</li> <li>– NO ACM are set in faulty/cut-out status</li> <li>– NO Towing mode is active</li> <li>– NO Fire alarm active on the train</li> <li>– Propulsion system ready ("Status: <i>Preheating possible</i>" signal is true)</li> </ul> | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1194 | PCU shall provide to TCMS interface signal to manage pre-conditioning mode <ul style="list-style-type: none"> <li>– Status: <i>Preheating possible</i></li> <li>– Command: <i>Activate Preheating</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement   | 4S_05.-.Propulsion          |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1195 | If all following condition are <u>not verified</u> , PCU shall set true "Status: <i>Preheating possible</i> " signal: <ul style="list-style-type: none"> <li>– No communication with the LIM</li> <li>– No communication with remote high voltage</li> <li>– DC system is isolated</li> <li>– DC pantograph is isolated</li> <li>– Any DC system disconnecter failure is active</li> <li>– Line trip global relay cannot open (all other line trip related failures are covered above)</li> </ul>                                                                                                       | Derived Requirement   | 4S_05.-.Propulsion          |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1196 | If Driver has pressed PC soft key on IDU, TCMS shall: <ul style="list-style-type: none"> <li>– set "Command: <i>Activate Preheating</i>" to all leader PCUs</li> <li>– activate traction block</li> <li>– Block panto raising command till both DC roof line disconnecter are open</li> </ul>                                                                                                                                                                                                                                                                                                           | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1197 | When "Command: <i>Activate Preheating</i> " signal is set, leader PCU shall open both DC line roof disconnecter and configure Line Trip chain in Local mode                                                                                                                                                                                                                                                                                                                                                                                                                                             | Derived Requirement   | 4S_05.-.Propulsion          |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1198 | In case one or both panto does not raise on order a warning to the driver shall be displayed that PC is not possible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1199 | In PC mode, with both panto up (and roof disconnecter open) when driver commands LCB closure, order shall be set in both half train                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1200 | In case one or both LCB does not close on order a warning to the driver shall be displayed that PC is not possible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A |

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| 2F_04.01.Zefiro-Europe.CONCEPT.1201 | When all ACM have started (in operation), TCMS shall activate " <i>Preconditioning mode request</i> " in all HVAC units                                                                                                          | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1202 | If within 60 sec from LCB closure, ALL ACM are not in operation status, LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively) | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1203 | If AC mode is selected, pre-conditioning mode has to be managed automatically by TCMS (according HVAC SFRS - 2F_15.02.Zefiro-Europe.TRS.61)                                                                                      | Derived Requirement | 4S_09.03.05.-.TCMS Software |       | N/A |

### Abort pre-conditioning mode

| Nr                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Applicability Details | Derived to                  | User Interface | SIL |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|----------------|-----|
| 2F_04.01.Zefiro-Europe.CONCEPT.1204 | An advice " <i>preconditioning finished</i> " shall be displayed to the driver when all HVAC go out of " <i>Preconditioning</i> " status                                                                                                                                                                                                                                                                                                                                                                      | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1205 | If driver presses Pre-conditioning softkey, when PC mode is active, " <i>Preconditioning mode</i> " shall be reset to HVAC (" <i>Automatic mode</i> " will be enabled), then request to open LCB and lower panto has to be displayed on IDU                                                                                                                                                                                                                                                                   | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1206 | When both panto are lowered after driver stop PC mode via IDU, TCMS will reset " <i>Command: Activate Preheating</i> "                                                                                                                                                                                                                                                                                                                                                                                        | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1207 | When " <i>Command: Activate Preheating</i> " signal is reset by TCMS, leader PCU shall close both DC line roof disconnect and reset Line trip chain configuration                                                                                                                                                                                                                                                                                                                                             | Derived Requirement   | 4S_05.-.Propulsion          |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1208 | In case any precondition is missing (except " <i>Both pantographs are down</i> "), TCMS shall <ul style="list-style-type: none"> <li>open LCB and reset "<i>Preconditioning mode request</i>" signal to all HVAC units</li> <li>display an warning to the driver that pre-conditioning is not possible</li> <li>after LCB is open request to lower pantographs shall be displayed to the driver and when driver lowers panto "<i>Command: Activate Preheating</i>" shall be set LOW to leader PCUs</li> </ul> | Derived Requirement   | 4S_09.03.05.-.TCMS Software |                | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1209 | In case during PC mode, one (or both) LCB is opened by local protection (but <i>Status: Preheating possible</i> signal is still set), second LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)                                                                                                                                                                                                         | Derived Requirement   | 4S_09.03.05.-.TCMS Software | EVENT          | N/A |

|                                    |                         |                         |                     |                        |
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| 2F_04.01.Zefiro-Europe.CONCEPT.1210 | <p>In case during PC mode, Line Voltage <math>\leq 2,2</math> kV DC on 3kV lines (or <math>\leq 1,4</math> kV on 1,5kV lines), if it doesn't rise over the threshold within 30s TCMS shall open all LCB</p> <p>(all parameters, time and voltage values, shall be tunable during commissioning)</p> <p>When line voltage comes back and it is over the threshold for more than 30s, LCB shall be automatically closed and pre-conditioning can start again (TCMS shall wait that ALL ACM are in operation and then sent "<i>Preconditioning mode request</i>" to all HVAC)</p> <p>(all parameters, time and voltage values, shall be tunable during commissioning)</p> <p>TCMS shall set a warning to the driver describing that pre-conditioning is not possible</p> | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1211 | <p>In case during PC mode, 1 ACM is stopped/faulty and it does not restart within 30s, all LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_04.01.Zefiro-Europe.CONCEPT.1212 | <p>In case during PC mode, at least 2 ACM are stopped/faulty, all LCB shall be immediately opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |

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## 4 Signal Interface

### 4.1 I/O Interface

#### 4.1.1 Supplier system

**Inputs:**

**Outputs:**

#### 4.1.2 TCMS SW

**Inputs:**

| Nr                                  | Functional name                         | Type | Linked to requirement                                                                                                                                |
|-------------------------------------|-----------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.30   | Pantograph Raised                       | L    | 2F_04.01.Zefiro-Europe.CONCEPT.284<br>2F_04.01.Zefiro-Europe.CONCEPT.309<br>2F_04.01.Zefiro-Europe.CONCEPT.310<br>2F_04.01.Zefiro-Europe.CONCEPT.484 |
| 2F_04.01.Zefiro-Europe.CONCEPT.587  | Pantograph Lowered                      | L    | 2F_04.01.Zefiro-Europe.CONCEPT.207<br>2F_04.01.Zefiro-Europe.CONCEPT.284                                                                             |
| 2F_04.01.Zefiro-Europe.CONCEPT.590  | Pantograph disconnecter open            | L    | 2F_04.01.Zefiro-Europe.CONCEPT.566<br>2F_04.01.Zefiro-Europe.CONCEPT.567                                                                             |
| 2F_04.01.Zefiro-Europe.CONCEPT.1077 | Pantograph disconnecter closed          |      | 2F_04.01.Zefiro-Europe.CONCEPT.566<br>2F_04.01.Zefiro-Europe.CONCEPT.567                                                                             |
| 2F_04.01.Zefiro-Europe.CONCEPT.604  | A-key activated                         | L    | 2F_04.01.Zefiro-Europe.CONCEPT.602                                                                                                                   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1044 | Line voltage selector: AC 25 kV 50 Hz   |      | 2F_04.01.Zefiro-Europe.CONCEPT.492                                                                                                                   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1045 | Line voltage selector: AC 15 kV 16.7 Hz |      | 2F_04.01.Zefiro-Europe.CONCEPT.492                                                                                                                   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1046 | Line voltage selector: 3 kV DC          |      | 2F_04.01.Zefiro-Europe.CONCEPT.492                                                                                                                   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1047 | Line voltage selector: 1.5 kV DC        |      | 2F_04.01.Zefiro-Europe.CONCEPT.492                                                                                                                   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1048 | LCB switch: LCB close                   |      | 2F_04.01.Zefiro-Europe.CONCEPT.374                                                                                                                   |
| 2F_04.01.Zefiro-Europe.CONCEPT.1049 | LCB switch: LCB open                    |      | 2F_04.01.Zefiro-Europe.CONCEPT.374                                                                                                                   |

PCU major fault

**Outputs:**

Pantograph raise order

Pantograph raise order backup mode

| Nr                                 | Functional name                  | Type | Linked to requirement              |
|------------------------------------|----------------------------------|------|------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.597 | Fast lowering of all pantographs | L    | 2F_04.01.Zefiro-Europe.CONCEPT.296 |

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| 2F_04.01.Zefiro-Europe.CONCEPT.589  | Indication lamp: LCB is open           | L | 2F_04.01.Zefiro-Europe.CONCEPT.985<br>2F_04.01.Zefiro-Europe.CONCEPT.986 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1075 | Command: Close pantograph disconnecter |   | 2F_04.01.Zefiro-Europe.CONCEPT.563                                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.1076 | Command: Open pantograph disconnecter  |   | 2F_04.01.Zefiro-Europe.CONCEPT.565                                       |

## 4.2 Bus Interface

Supplier to TCMS SW: PCU -> CCU-O

| Nr                                  | Functional name                                                      | Type | Linked to requirement                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.901  | Command: enable start of remote 1500V DC system                      |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.902  | Command: enable start of 1500V DC system                             |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.903  | Command: enable start of remote 3kV DC system                        |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.904  | Command: enable start of 3kV DC system                               |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.905  | Command: enable start of remote 15kV AC system                       |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.906  | Command: enable start of 15kV AC system                              |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.907  | Command: enable start of remote 25kV AC system                       |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.908  | Command: enable start of 25kV AC system                              |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.910  | Command: open line circuit breaker                                   |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.911  | Status: Line trip                                                    |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.913  | Command: isolate local AC system                                     |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.914  | Max traction power due to transformer supervision                    |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.915  | Max line current RMS                                                 |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.916  | Command: AC line regeneration inhibit due to transformer supervision |      |                                                                                                                                                      |
| 2F_04.01.Zefiro-Europe.CONCEPT.1106 | Command: Isolate pantograph                                          |      | 2F_04.01.Zefiro-Europe.CONCEPT.531<br>2F_04.01.Zefiro-Europe.CONCEPT.762<br>2F_04.01.Zefiro-Europe.CONCEPT.763<br>2F_04.01.Zefiro-Europe.CONCEPT.995 |

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**TCMS SW to Supplier: CCU-O -> PCU**

| Nr                                 | Functional name                                           | Type | Linked to requirement |
|------------------------------------|-----------------------------------------------------------|------|-----------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.917 | Country Code                                              |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.918 | Status: neutral section                                   |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.919 | Status: separation section                                |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.934 | Status: enable high voltage control from TCMS             |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.935 | Status: Remote DC disconnecter is closed                  |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.936 | Command: Enable start of 1500V DC from remote system      |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.937 | Status: Remote DC high voltage system is isolated         |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.938 | Command: Enable start of 1500V DC system from master PCU  |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.920 | Status: Remote DC disconnecter is closed                  |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.921 | Command: enable start of 3kV DC system from remote system |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.922 | Status: Remote DC high voltage system is isolated         |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.923 | Command: Enable start of 3kV DC system from master PCU    |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.924 | Status: Remote AC disconnecter is closed                  |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.925 | Command: Enable start of 15kV AC from remote system       |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.926 | Command: Enable start of 15kV AC system from master PCU   |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.927 | Status: Remote AC disconnecter switch is closed           |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.928 | Command: Enable start of 25kV AC from remote system       |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.929 | Command: Enable start of 25kV AC system from master PCU   |      |                       |
| 2F_04.01.Zefiro-Europe.CONCEPT.909 | Status:All pantographs down                               |      |                       |

Status: Line circuit breaker closed

| Nr                                 | Functional name                                                                          | Type | Linked to requirement |
|------------------------------------|------------------------------------------------------------------------------------------|------|-----------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.930 | Max traction power due to transformer supervision from redundant PCU on same transformer |      |                       |

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|------------------------------------|-------------------------------------------------------------------------------------------------------------|--|--|
| 2F_04.01.Zefiro-Europe.CONCEPT.931 | Max line circuit current from redundant PCU on same transformer, RMS                                        |  |  |
| 2F_04.01.Zefiro-Europe.CONCEPT.932 | Valid bit:communication with redundant PCU on same transformer                                              |  |  |
| 2F_04.01.Zefiro-Europe.CONCEPT.933 | Command: AC line regeneration inhibit due to transformer supervision from redundant PCU on same transformer |  |  |

**Supplier to TCMS SW: Panto -> CCU-O**

| Nr                                 | Functional name                        | Type | Linked to requirement                                                      |
|------------------------------------|----------------------------------------|------|----------------------------------------------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.942 | Pantograph 1 position                  |      | 2F_04.01.Zefiro-Europe.CONCEPT.284                                         |
| 2F_04.01.Zefiro-Europe.CONCEPT.943 | Pantograph 2 position                  |      | 2F_04.01.Zefiro-Europe.CONCEPT.284                                         |
| 2F_04.01.Zefiro-Europe.CONCEPT.944 | Pantograph 1 unintentionally raised    |      |                                                                            |
| 2F_04.01.Zefiro-Europe.CONCEPT.945 | Pantograph 2 unintentionally raised    |      |                                                                            |
| 2F_04.01.Zefiro-Europe.CONCEPT.947 | Pantograph 1 unintentionally lowered   |      | 2F_04.01.Zefiro-Europe.CONCEPT.207<br>2F_04.01.Zefiro-Europe.CONCEPT.309   |
| 2F_04.01.Zefiro-Europe.CONCEPT.948 | Pantograph 2 unintentionally lowered   |      | 2F_04.01.Zefiro-Europe.CONCEPT.207<br>2F_04.01.Zefiro-Europe.CONCEPT.309   |
| 2F_04.01.Zefiro-Europe.CONCEPT.949 | Pantograph 1 does not raise on command |      | 2F_04.01.Zefiro-Europe.CONCEPT.207<br>2F_04.01.Zefiro-Europe.CONCEPT.309   |
| 2F_04.01.Zefiro-Europe.CONCEPT.950 | Pantograph 2 does not raise on command |      | 2F_04.01.Zefiro-Europe.CONCEPT.207<br>2F_04.01.Zefiro-Europe.CONCEPT.309   |
| 2F_04.01.Zefiro-Europe.CONCEPT.951 | Pantograph 1 does not lower on command |      |                                                                            |
| 2F_04.01.Zefiro-Europe.CONCEPT.952 | Pantograph 2 does not lower on command |      |                                                                            |
| 2F_04.01.Zefiro-Europe.CONCEPT.953 | Pantograph 1 rise time too long        |      |                                                                            |
| 2F_04.01.Zefiro-Europe.CONCEPT.954 | Pantograph 2 rise time too long        |      |                                                                            |
| 2F_04.01.Zefiro-Europe.CONCEPT.955 | Pantograph 1 ADD tripped               |      | 2F_04.01.Zefiro-Europe.CONCEPT.294<br>2F_04.01.Zefiro-Europe.CONCEPT.332   |
| 2F_04.01.Zefiro-Europe.CONCEPT.956 | Pantograph 2 ADD tripped               |      | 2F_04.01.Zefiro-Europe.CONCEPT.294<br>2F_04.01.Zefiro-Europe.CONCEPT.332   |
| 2F_04.01.Zefiro-Europe.CONCEPT.957 | Pantograph 1 wear detection            |      | 2F_04.01.Zefiro-Europe.CONCEPT.1062<br>2F_04.01.Zefiro-Europe.CONCEPT.1063 |
| 2F_04.01.Zefiro-Europe.CONCEPT.958 | Pantograph 2 wear detection            |      | 2F_04.01.Zefiro-Europe.CONCEPT.1061<br>2F_04.01.Zefiro-Europe.CONCEPT.1062 |

|                                    |                         |                         |                     |                        |
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**TCMS SW to Supplier: CCU-O -> Panto**

| Nr                                  | Functional name                           | Type | Linked to requirement               |
|-------------------------------------|-------------------------------------------|------|-------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.960  | Train speed                               |      | 2F_04.01.Zefiro-Europe.CONCEPT.97   |
| 2F_04.01.Zefiro-Europe.CONCEPT.961  | Selected Voltage                          |      | 2F_04.01.Zefiro-Europe.CONCEPT.97   |
| 2F_04.01.Zefiro-Europe.CONCEPT.962  | Selected catenary                         |      | 2F_04.01.Zefiro-Europe.CONCEPT.97   |
| 2F_04.01.Zefiro-Europe.CONCEPT.963  | Pantograph position command (raise order) |      | 2F_04.01.Zefiro-Europe.CONCEPT.283  |
| 2F_04.01.Zefiro-Europe.CONCEPT.1120 | Wear detection blocked (ICD Weardetec)    |      | 2F_04.01.Zefiro-Europe.CONCEPT.1119 |

**Supplier to TCMS SW: EMS -> CCU-O**

| Nr                                  | Functional name                      | Type | Linked to requirement               |
|-------------------------------------|--------------------------------------|------|-------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1321 | Diagnostics                          |      | 2F_04.01.Zefiro-Europe.CONCEPT.1319 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1322 | Energy data (for ground station DCS) |      | 2F_04.01.Zefiro-Europe.CONCEPT.1315 |

**TCMS SW to Supplier: CCU-O -> EMS**

| Nr                                  | Functional name     | Type | Linked to requirement               |
|-------------------------------------|---------------------|------|-------------------------------------|
| 2F_04.01.Zefiro-Europe.CONCEPT.1324 | location (GPS) data |      | 2F_04.01.Zefiro-Europe.CONCEPT.1313 |
| 2F_04.01.Zefiro-Europe.CONCEPT.1325 | time/date data      |      | 2F_04.01.Zefiro-Europe.CONCEPT.1313 |

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## 5 References

[1] Data Interface (ICD) Energy Metering System VZES 3EGH489028-5141

|                             |                  |                  |              |                        |
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## 6 Revision Log

### 6.1 Revision log 3EGH489028-2358\_

|                        |                                 |              |                                                                                                                                  |
|------------------------|---------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Revision:</b>       | —                               |              |                                                                                                                                  |
| <b>Date:</b>           | 2020-11-09                      |              |                                                                                                                                  |
| <b>Editor:</b>         | S Dhiraj                        |              |                                                                                                                                  |
| <b>Change ID DOORS</b> | <b>Refers to requirement ID</b> | <b>Title</b> | <b>Change content</b>                                                                                                            |
|                        |                                 |              | First Issue<br>The document derives from 3EST000225-7907_C of V300ZEFIRO Italy modified with CR 60348 and CR 61490 and CR 177056 |

### 6.2 Revision log 3EGH489028-2358\_A

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Revision:</b>                        | _A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Date:</b>                            | 2021-02-18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Editor:</b>                          | M Ciabatti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Change ID DOORS</b>                  | <b>Refers to requirement ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Title</b>        | <b>Change content</b>                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2F_04.01.ISSUES.-<br>.Zefiro-Europe.424 | [2F_04.01.Zefiro-Europe.CONCEPT.1290]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1291]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1292]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1293]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1295]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1296]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1297]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1298]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1300]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1301]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1302]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1303]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1304]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1305]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1306]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1307]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1333]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1339]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1340]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1341]: | ADD bypass - VZI 14 | CR 35133<br>In case of unavailability of both pantographs of the available system due to ADD intervention (or failure), the train is stopped on the main line.<br>This change will introduce the possibility to move the train to reach a siding track to clear the main line: through a sealed switch in driver's cab the driver, under his responsibility, can allow to raise the pantograph and move the train at reduced speed. |

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### 6.3 Revision log 3EGH489028-2358\_B

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Revision:</b>                        | <b>_B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Date:</b>                            | <b>2021-05-25</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Editor:</b>                          | <b>M Ciabatti</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Change ID DOORS</b>                  | <b>Refers to requirement ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Title</b>                                                        | <b>Change content</b>                                                                                                                                                                                                                                                                                                                                                                                           |
| 2F_04.01.ISSUES.-<br>.Zefiro-Europe.426 | [2F_04.01.Zefiro-Europe.CONCEPT.1342]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1343]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1344]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1345]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1347]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1348]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1349]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1350]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1352]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1353]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1354]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1355]:<br>[2F_04.01.Zefiro-Europe.TRS.798]:<br>[2F_04.01.Zefiro-Europe.TRS.799]: | VZES - Functional:<br>Integration of Energy Measurement System Data | CR 200970<br><br>DC data will come to the Energy Measurement Device by ETH bus.<br>AC data will come to Energy Measurement Device by electrical connections. Due to limited inputs of Energy Measurement Device the pantograph to be monitored shall be selected by TCMS.<br>Energy data from Energy Measurement Device is sent to the wayside by MCG.<br>Status of Energy Measurement Device is showed on IDU. |
| 2F_04.01.ISSUES.-<br>.Zefiro-Europe.427 | [2F_04.01.Zefiro-Europe.CONCEPT.1345]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.1354]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.6]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.775]:<br>[2F_04.01.Zefiro-Europe.CONCEPT.965]:<br>[2F_04.01.Zefiro-Europe.TRS.772]:                                                                                                                                                                                                                                                                                                                                                 | VZES: Functional:<br>country code management by TCMS                | CR 204784<br><br>Alignment the REQ to the management of the country code implemented on V300ZEFIRO.<br><br>Note: added also some editorial updates before release the document in PDM.                                                                                                                                                                                                                          |